

Does the Bias of Empirical Tail Value-at-Risk Matter in Practice?

An Asymptotic Expansion Perspective

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Abstract

Tail Value-at-Risk (TVaR) is a widely adopted risk measure playing a critically important role in both academic research and industry practice in insurance. In data applications, TVaR is often estimated using the empirical method, owing to its simplicity and nonparametric nature. The empirical TVaR has been explicitly advocated by regulatory authorities as a standard approach for computing TVaR. However, prior literature has pointed out that the empirical TVaR estimator is negatively biased, which can lead to a systematic underestimation of risk in finite-sample applications. This paper aims to deepen the understanding of the bias of the empirical TVaR estimator in two dimensions: its magnitude as well as the key distributional and structural determinants driving the severity of the bias. To this end, we derive a leading-term approximation for the bias based on its asymptotic expansion. The closed-form expression associated with the leading-term approximation enables us to obtain analytical insights into the structural properties governing the bias of the empirical TVaR estimator. To account for the discrepancy between the leading-term approximation and the true bias, we further derive an explicit bound for the bias. We validate the proposed bias analysis framework via simulations. Two real datasets are used to demonstrate the practical relevance of the framework, each representing a distinct setting where the bias may or may not be negligible.

Keywords: Expected shortfall, bootstrap, finite-sample bias, heavy-tailed distributions, trimmed means

Dedication:

This research was initiated by Ričardas Zitikis (1963–2025) in early 2025. This article is dedicated to his memory.

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1 Introduction

The measurement of risk plays a pivotal role across a wide spectrum of applications in finance and insurance, including but not limited to, determining risk-loaded prices, assessing capital adequacy, supporting regulatory compliance, measuring risk-based performance, and guiding portfolio optimization. Among many risk measures employed in these contexts, Tail Value-at-Risk (TVaR), which is also referred to as AVaR, CVaR, or expected shortfall, has gained particular prominence among academics and professionals, especially following its adoption as the replacement for Value-at-Risk (VaR) as the standard risk measure within recent regulatory frameworks (e.g., [BCBS, 2016](#), [2019](#)). In detail, we consider the risk position of a generic one-period model, denoted by X , where positive values represent losses and negative values represent surpluses. Let F be the cumulative distribution function (CDF) of X . The left-continuous inversion of F is given by

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}, \quad u \in (0, 1).$$

For a given probability level $p \in (0, 1)$, the TVaR of X at level p is defined, provided the integral is finite, as

$$T_p = \frac{1}{1-p} \int_p^1 F^{-1}(u) \, du,$$

which can be interpreted as the average of the upper tail quantiles of X .

Given the practical prominence of TVaR, a large and continually growing body of literature has been generated. A wide range of its economically grounded properties have been extensively studied, including its adherence to the coherence axioms ([Artzner et al., 1999](#)), comonotonic additivity ([Kusuoka, 2001](#)), and its representation as a Choquet integral ([Schmeidler, 1989](#)) and as the solution to an expected loss minimization problem ([Rockafellar and Uryasev, 2002](#)). Moreover, we refer readers to [Wang and Zitikis \(2021\)](#) for an axiomatic characterization of TVaR, which also includes a comprehensive review of recent theoretical developments of the notion.

While different from, yet closely related to, the aforementioned literature on studying the mathematical and economic properties of TVaR, the primary objective of this paper pertains to the estimation of TVaR. In applied settings, the choice of estimation method is often guided by practical considerations. Practitioners often favor approaches that are simple to implement, transparent to communicate, and whose performance does not rely on a specific set of parametric assumptions. One such widely adopted method is the empirical, or plug-in, estimator of TVaR:

$$\hat{T}_{n,p} = \frac{1}{1-p} \int_p^1 \hat{F}_n^{-1}(u) \, du, \tag{1.1}$$

where $\widehat{F}_n^{-1}$ denotes the inversion of the empirical CDF $\widehat{F}_n$, based on $n \in \mathbb{N}$ independent and identically distributed (iid) observations $X_1, \dots, X_n$ drawn from the distribution of X . In particular, the empirical CDF is computed via

$$\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i \leq x\}},$$

where $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function, and its inversion is calculated as

$$\widehat{F}_n^{-1}(u) = \inf\{x \in \mathbb{R} : \widehat{F}_n(x) \geq u\} = \begin{cases} X_{nu:n}, & \text{if } nu \text{ is an integer;} \\ X_{[nu]+1:n}, & \text{otherwise,} \end{cases} \quad u \in (0, 1),$$

where $[\cdot]$ represents the greatest integer function, and $X_{1:n} \leq \dots \leq X_{n:n}$ are the order statistics of the samples $X_1, \dots, X_n$. Throughout the rest of this paper, empirical TVaR refers to the estimator in (1.1). This empirical estimator has been recommended by the European Banking Authority as the standard formula for computing TVaR (EBA, 2020).

The large-sample properties of the empirical TVaR $\widehat{T}_{n,p}$ have been thoroughly studied in the related literature (e.g., Brazauskas et al., 2008; Wei and Zitikis, 2023). More general results based on the empirical tail conditional allocation were established in Gribkova et al. (2022a,b). In particular, it has been shown that $\widehat{T}_{n,p}$ in (1.1) is consistent. Hence, the empirical TVaR $\widehat{T}_{n,p}$ ultimately yields an accurate evaluation of the true TVaR as the sample size tends to infinity. That said, in real-world applications, data are finite and sometimes rather limited. In the finite-sample setting, the empirical TVaR estimator in (1.1) has been found to exhibit a negative bias (Wei and Zitikis, 2023; Gribkova et al., 2026). Notably, this concern is not merely of academic interest. Practitioners have also arrived at a similar conclusion through numerical experiments (García-Risueño, 2025).

Existing literature has established the existence and sign of the finite-sample bias of the empirical TVaR estimator. However, to the best of our knowledge, no prior study has thoroughly addressed the severity of the bias. Consequently, two possible scenarios emerge for users of the empirical TVaR. If the bias is small relative to the value of TVaR, then the issue is not a major practical concern, and the empirical TVaR estimator can be used with confidence. If instead the bias is substantial, then caution is warranted, and further action such as implementing a bias reduction strategy, may be necessary. Otherwise, the inherent negative bias may lead to a systematic underestimation of the riskiness of a financial position as measured by TVaR.

The goal of our paper is to examine the magnitude of the negative bias in the empirical TVaR estimator $\widehat{T}_{n,p}$, thus providing insights into the two potential scenarios discussed above. Note that our paper does not intend to simply put forth a bias-reduced estimator for the reader to adopt as

a matter of course. Instead, we seek to understand a more fundamental question:

Q1. Does the bias of the empirical TVaR matter in actuarial practice, and if so, under what conditions?

While numerical methods such as the jackknife (Quenouille, 1956; Tukey, 1958) and the bootstrap (Efron and Tibshirani, 1994) can be used to estimate the bias of an estimator for a given dataset, the resulting bias estimates are inherently tied to the specific sample at hand. As such, these methods are limited in their ability to provide a structural understanding of how the distributional characteristics of the data affect the extent of the bias. Thus, they fall short of answering the “under what conditions” part of question Q1 in an insightful way. Moreover, as we will see in the real-data analysis, numerical bias evaluation methods may, due to sampling variability, occasionally yield improperly positive bias estimates for the empirical TVaR, contradicting the fact that the bias is provably negative.

In this paper, we take an analytical route to analyze the bias of the empirical TVaR estimator in (1.1). Specifically, following the initial conjecture the same authors of the present paper made in equation (30) of Gribkova et al. (2026), this paper formally establishes the asymptotic expansion of the bias under detailed conditions and with a rigorous proof. We then use the leading term of the expansion as a practical approximation for analyzing the bias. It is worth emphasizing that, while the leading term is identified based on the decay rate of the individual constituent components of the bias as $n \rightarrow \infty$, its practical utility is by no means confined to the infinite-sample regime. Speaking plainly, the leading-term approximation can be interpreted as the statistical analogue of a standard strategy in quantitative risk management, where a complex risk position is decomposed into material and immaterial risk factors to make its analysis more tractable. In a parallel fashion, the leading term of the bias expansion represents the most material driver of the bias, whose materiality is defined by its slower rate of diminishment as n increases, relative to the higher-order remainder terms. As will be seen in Section 4, the leading-term approximation is able to capture the true value of the bias accurately across a wild range of finite-sample experiments.

The aforementioned leading-term approximation, while informative, does not coincide exactly with the actual bias of the empirical TVaR estimator. We therefore supplement it with an explicit bound. Our numerical experiments indicate that, with an appropriate choice of tuning parameters, the bound can achieve a reasonable balance between tightness and conservativeness for assessing the bias.

Our paper contains both theoretical and practical contributions. On the theoretical front, we derive an asymptotic expansion, together with an explicit bound, for the bias of the empirical TVaR, under minimal conditions. It is noteworthy that a standard approach to such an expansion typically proceeds by directly invoking existing results on the expansion of the empirical quantile function

(see [Bahadur, 1966](#); [Kiefer, 1970](#), for the Bahadur–Kiefer asymptotic expansion). Adopting this standard approach would require the underlying density to be differentiable throughout the entire tail region. However, it is a stylized fact that insurance loss data are sometimes subject to an upper limit, for example due to a policy limit or a stop-loss reinsurance treaty. This gives rise to an atom in the loss distribution at the corresponding limit, causing the required differentiability condition to fail. To maximize the applicability of the bias analysis result, in this paper we instead pursue a more technically demanding approach that involves decomposing the estimation error of the empirical TVaR into a sum of centered winsorized random variables and an explicit remainder term. This approach allows us to establish the bias expansion under only the continuity of the density in a neighborhood of the p -th quantile, rather than differentiability across the entire tail, therefore significantly expanding the range of settings in which the bias analysis framework holds true.

On the practical front, the bias analysis in this paper provides an analytical and diagnostic insight into the severity of the bias of the empirical TVaR. More specifically, our analysis shows that the answer to question [Q1](#) is, in fact, two-sided. In a data-sufficient setting where the sample size is at least moderate, the bias is small and perhaps negligible, even in the presence of heavy-tailed distributions. However, a contrasting, cautionary regime is characterized by constrained sample sizes, heavy-tailed losses, and a high probability level p for computing TVaR. In this regime, the bias can be a substantial fraction of the value of TVaR, and thus is non-negligible. Under these circumstances, the leading-term approximation can readily serve to reduce the bias, and the bias bound can be used as an additional safety buffer. As a side note, we remark that it is not uncommon for actuaries to operate and form judgments based on limited data. Familiar circumstances include large claims management in reinsurance business, financial and natural catastrophe modeling, and the study of emerging risks such as operational, cyber, or AI-induced losses.

We note that, in addition to the empirical method considered in this paper, other commonly used approaches for estimating TVaR include parametric methods (e.g., [Chen et al., 2023](#); [Brazauskas and Kleefeld, 2009](#); [Lee and Lin, 2010](#)), kernel-based methods (e.g., [Bolance et al., 2008](#); [Chen, 2008](#); [Yu et al., 2010](#)), and methods originating from Extreme Value Theory (e.g., [Goegebeur et al., 2022](#); [Hua and Joe, 2011](#); [Necir et al., 2010](#)). We also refer the reader to [Nadarajah et al. \(2014\)](#) for a systematic review of different methods for estimating TVaR. It is important to note that the objective of this paper is not to advocate for the superiority of one estimation method over another for computing TVaR. Rather, given the widespread use of the empirical TVaR estimator in [\(1.1\)](#), this paper contributes to deepening the fundamental understanding of its properties and limitations, and explores how it may be improved when necessary.

Finally, let us explain the practical considerations behind our choice to focus on studying the

bias of the empirical TVaR estimator, rather than its variance, which is another equally essential component in quantifying estimation errors (see Theorem 3.3 in [Gribkova et al., 2022b](#), for the variance analysis for the empirical TVaR). The first consideration pertains to the systematic nature of the error captured by bias. The type of error captured by variance is idiosyncratic. When the same risk measure estimator is applied repeatedly, either through periodic risk assessments of a company’s portfolio (e.g., quarterly or annual assessments), or by a regulator for assessing the riskiness of multiple business lines or market participants, the upward and downward idiosyncratic errors tend to cancel out across these repetitions, so that the estimates remain useful for informing the long-term or system-wide tendency of the risk landscape. Bias, by contrast, represents a systematic error. It corresponds to a structural shortfall of the estimator, and the resulting error is persistent and cannot be averaged out even across repeated applications, skewing every application in the same direction.

The other consideration pertains to the asymmetric consequences of underestimation and overestimation of a risk quantity. When TVaR is used to determine the economic capital needed for an insurance position, underestimation implies undercapitalization, which can endanger the solvency of the insurer, whereas overestimation, though resulting in overcapitalization, only incurs an additional capital cost and reduces profitability. Variance does not differentiate between the impacts of the two directions of error, but bias reflects a systematic shift of the estimation in one particular direction. In the case of the empirical TVaR estimator, the bias is always negative, implying the more consequential case of underestimation, or equivalently undercapitalization.

The remainder of this paper is organized as follows. Before presenting the main results, Section 2 compares the bias of the empirical TVaR estimator in (1.1) with that of an alternative empirical estimator based on the conditional-mean formulation. Section 3 then presents the proposed explicit bias analysis framework. Section 4 assesses the performance of the bias quantification framework through extensive simulations, with additional robustness check experiments included in Appendix A. In Section 5, the practical relevance of the proposed framework is illustrated using two real datasets, each representing a distinct setting where the bias may or may not be negligible. Section 6 concludes the paper. To facilitate readability, all technical proofs are relegated to Appendix B.

2 A commentary on empirical Tail Conditional Expectation

Before presenting the bias analysis framework, this section is devoted to comparing the empirical TVaR estimator $\hat{T}_{n,p}$ in (1.1), from the perspective of bias, with another empirical estimator based on the conditional-mean formulation.

In what follows, for notational convenience, let us shorthand the p -th left percentile of F by

$$\xi_p := F^{-1}(p), \quad p \in (0, 1).$$

The following relationship holds:

$$T_p = \frac{1}{1-p} \int_p^1 F^{-1}(u) \, du = \frac{1}{1-p} \left[\mathbf{E} \left(X \mathbf{1}_{\{X > \xi_p\}} \right) + \xi_p \times \left(F(\xi_p) - p \right) \right].$$

Moreover, when the quantile function F is continuous around ξ_p , it holds that $F(\xi_p) = p$, and the above expression simplifies to

$$T_p = \mathbf{E}(X \mid X > \xi_p),$$

where the right-hand side corresponds to the Tail Conditional Expectation (TCE) risk functional. In this case, TVaR can be estimated using the empirical TCE estimator, defined as

$$\widehat{\text{TCE}}_{n,p} = \frac{1}{n - [np]} \sum_{i=[np]+1}^n X_{i:n}. \quad (2.1)$$

We assert that the empirical TVaR estimator of interest, namely $\widehat{T}_{n,p}$ in (1.1), exhibits a smaller magnitude of negative bias (i.e., a smaller absolute bias) than $\widehat{\text{TCE}}_{n,p}$ in (2.1), regardless of whether the distribution F is continuous or not. Therefore, from the perspective of bias minimization, $\widehat{T}_{n,p}$ in (1.1) should be preferred over $\widehat{\text{TCE}}_{n,p}$ in (2.1) for estimating TVaR. To show the above assertion, let us rewrite the empirical TVaR estimator in terms of the following more computationally friendly form:

$$\begin{aligned} \widehat{T}_{n,p} &= \frac{1}{1-p} \int_p^1 \widehat{F}_n^{-1}(u) \, du = \frac{1}{1-p} \left[\int_p^{\frac{[np]+1}{n}} \widehat{F}_n^{-1}(u) \, du + \int_{\frac{[np]+1}{n}}^1 \widehat{F}_n^{-1}(u) \, du \right] \\ &= \frac{[np] + 1 - np}{(1-p)n} X_{[np]+1:n} + \frac{1}{(1-p)n} \sum_{i=[np]+2}^n X_{i:n} \\ &= -\frac{np - [np]}{(1-p)n} X_{[np]+1:n} + \frac{1}{(1-p)n} \sum_{i=[np]+1}^n X_{i:n} \\ &= -\frac{np - [np]}{(1-p)n} X_{[np]+1:n} + \frac{n - [np]}{(1-p)n} \widehat{\text{TCE}}_{n,p}. \end{aligned} \quad (2.2)$$

Accordingly, we have

$$\widehat{\text{TCE}}_{n,p} = \frac{(1-p)n}{n - [np]} \widehat{T}_{n,p} + \frac{np - [np]}{n - [np]} X_{[np]+1:n}$$

$$= \widehat{T}_{n,p} - \frac{np - [np]}{n - [np]} \left(\widehat{T}_{n,p} - X_{[np]+1:n} \right). \quad (2.3)$$

Recall that the bias of $\widehat{T}_{n,p}$ is always negative (Wei and Zitikis, 2023; Gribkova et al., 2026). Moreover, note that

$$\widehat{T}_{n,p} = \frac{1}{1-p} \int_p^1 \widehat{F}_n^{-1}(u) du \geq \frac{1}{1-p} \int_p^1 \widehat{F}_n^{-1}(p) du = \widehat{F}_n^{-1}(p).$$

If np is not an integer, then $\widehat{F}_n^{-1}(p) = X_{[np]+1:n}$, hence $\widehat{T}_{n,p} \geq X_{[np]+1:n}$. Therefore, the subtraction term in (2.3) is always non-negative, which introduces an additional negative bias to the estimator $\widehat{\text{TCE}}_{n,p}$ compared to $\widehat{T}_{n,p}$, unless np is an integer, in which case this additional bias is zero. In other words, the empirical estimator of interest $\widehat{T}_{n,p}$ always outperforms $\widehat{\text{TCE}}_{n,p}$ in the sense of having a smaller, or at the very least an equal, negative bias. This observation supports the regulatory choice of $\widehat{T}_{n,p}$ as the standard method for estimating TVaR over $\widehat{\text{TCE}}_{n,p}$, and it also justifies our focus on $\widehat{T}_{n,p}$ as the estimator of interest in this paper.

3 Main results

We aim to study the bias of the empirical TVaR $\widehat{T}_{n,p}$ in (1.1) under minimal assumptions, so that the results can be broadly applicable across a wide range of settings. Nevertheless, certain finiteness conditions are still required to guarantee that the problem under investigation is well-posed. The first condition ensures that the risk measure of interest, T_p , is well-defined:

$$(C_1) \quad \int_p^1 F^{-1}(u) du < \infty, \quad \text{for a given } p \in (0, 1).$$

In addition, we need the mean of the empirical TVaR estimator, and thus its bias, to exist. We let X_- denote the negative part of X . Ensuring the finiteness of the mean of $\widehat{T}_{n,p}$ needs the following extra condition:

$$(C_2) \quad \mathbf{E}(X_-^\varepsilon) < \infty, \quad \text{for some } \varepsilon > 0.$$

It is worth noting that condition (C₂) is very mild. In particular, it may fail only in the presence of a sufficiently heavy left tail. In the context of risk management, loss models are typically assumed to have positive support or at the very least lower bounded. In these cases, condition (C₂) is automatically satisfied.

To understand why conditions (C₁) and (C₂) together imply the finiteness of the bias, we shall examine the difference between $\widehat{T}_{n,p}$ and T_p . To this end, we need to introduce some additional notations. For $i = 1, \dots, n$, let W_i denote the winsorized counterpart of X_i with respect to the

threshold ξ_p :

$$W_i = \max(X_i, \xi_p) = \begin{cases} \xi_p, & X_i \leq \xi_p; \\ X_i, & X_i > \xi_p. \end{cases}$$

Note that $W_1, \dots, W_n$ are iid random variables. We denote their common mean by

$$\mu_W = \mathbf{E}(W_1).$$

Moreover, let $N_p := \text{card}\{i : X_i \leq \xi_p\}$ denote the number of iid samples $X_1, \dots, X_n$ that fall below or at the threshold ξ_p .

The following lemma spells out $(\hat{T}_{n,p} - T_p)$ explicitly in terms of a sum of iid centered winsorized random variables, accompanied by a non-positive remainder term. At the outset, let us define

$$a \wedge b := \min(a, b); \quad a \vee b := \max(a, b); \quad \text{sign}(a) := \begin{cases} \frac{a}{|a|}, & a \neq 0, \\ 0, & a = 0. \end{cases} \quad (3.1)$$

Lemma 3.1. *Suppose that condition $(\mathbf{C}_1)$ holds, then we have*

$$\hat{T}_{n,p} - T_p = \frac{1}{n(1-p)} \sum_{i=1}^n (W_i - \mu_W) - R_n, \quad (3.2)$$

where

$$R_n = \frac{np - [np]}{n(1-p)} (X_{[np]+1:n} - \xi_p) - \frac{\text{sign}(N_p - [np])}{n(1-p)} \sum_{i=(\lfloor np \rfloor \wedge N_p)+1}^{\lfloor np \rfloor \vee N_p} (X_{i:n} - \xi_p). \quad (3.3)$$

Furthermore, it holds almost surely that

$$R_n \geq 0. \quad (3.4)$$

From here on, let us denote the bias of the empirical TVaR $\hat{T}_{n,p}$ by

$$B_n = \mathbf{E}(\hat{T}_{n,p}) - T_p.$$

From the representation (3.2) in Lemma 3.1, it follows that

$$B_n = -\mathbf{E}(R_n). \quad (3.5)$$

Therefore, the finiteness of the bias B_n is equivalent to the finiteness of $\mathbf{E}(R_n)$, or equivalently, to

the finiteness of the expectations of the order statistics involved in the expression of R_n in (3.3). By Proposition 2 of [Stigler \(1974\)](#), conditions (C₁) and (C₂) together ensure the existence of moments of central order statistics of any order, and in particular, the finiteness of the first moment, which is sufficient for our purposes in the current context.

Remark 3.1. *By the relationships noted in (3.4) in Lemma 3.1 as well as in (3.5), it follows that the bias B_n is negative, which is in agreement with the result established in [Gribkova et al. \(2026\)](#).*

We are now in a position to present one of the main results of this paper.

Theorem 3.1. *For a fixed $p \in (0, 1)$, suppose that conditions (C₁) and (C₂) hold. Further, assume that the probability density function (PDF) of X , denoted by f , exists and is continuous in a neighborhood of the point $\xi_p = F^{-1}(p)$, and that $f(\xi_p) > 0$. Then, the following relationship holds for the bias of the empirical TVaR $\hat{T}_{n,p}$:*

$$B_n = \ell_n + o(n^{-1}), \quad \ell_n = -\frac{p}{2nf(\xi_p)}. \quad (3.6)$$

Remark 3.2. *In equation (30) of [Gribkova et al. \(2026\)](#), the same authors as in the present paper conjectured the asymptotic expansion in (3.6), motivated by heuristic reasoning derived from Edgeworth-type expansions for the distribution of normalized trimmed means studied in [Gribkova and Helmers \(2006, 2007\)](#). That conjecture was stated without a complete specification of the required conditions nor a rigorous proof. Theorem 3.1 of the present paper formally confirms the correctness of this initial conjecture by establishing the expansion under a complete and explicit set of conditions.*

Thanks to the explicit expression of the leading bias term established in Theorem 3.1, several analytical insights emerge immediately, which are summarized in the succeeding remarks.

Remark 3.3. *The asymptotic expansion in (3.6) shows that the negative bias decreases at the rate of n^{-1} as the sample size increases. It also confirms that the empirical TVaR estimator is asymptotically unbiased.*

Remark 3.4. *For a fixed sample size n , the leading term ℓ_n in (3.6) is proportional to $p/f(\xi_p)$. Assuming that f decreases monotonically along the far right tail, which is typical among insurance loss models, then the function $p \mapsto p/f(\xi_p)$ is increasing for sufficiently large values of p . This implies that the negative bias becomes more pronounced as the probability level p increases. Intuitively, increasing p reduces the size of the tail region and leads to fewer observations above the VaR threshold ξ_p , so the estimation of the tail mean becomes more difficult and its finite-sample bias becomes larger.*

Moreover, it is noted that the leading bias term ℓ_n explodes as $p \rightarrow 1$. This behavior coincides with that of the actual bias. Specifically, for a distribution with unbounded support, on the one hand, the TVaR risk measure T_p diverges as $p \rightarrow 1$. On the other hand, for a fixed n , once p is large enough (i.e., $p \geq (n-1)/n$), the empirical TVaR estimator collapses to the sample maximum, $\hat{T}_{n,p} = X_{n:n}$, whose mean is finite because of conditions (C₁) and (C₂). Consequently, the actual bias $B_n = \mathbf{E}(\hat{T}_{n,p}) - T_p$ diverges as $p \rightarrow 1$. This is a reassuring property, as it shows that the leading bias term captures the correct behavior of the actual bias even in the extreme limit as $p \rightarrow 1$.

Remark 3.5. When n and p are fixed, the magnitude of the leading bias term ℓ_n is inversely related to $f(\xi_p)$. In risk management practice, TVaR is commonly evaluated at high probability levels. When a data distribution becomes more heavy-tailed, its associated probability density tends to spread farther into the extreme region. As a result, the density around a fixed high quantile becomes smaller, or equivalently $f(\xi_p)$ tends to decrease. While this opposite relationship between tail heaviness and $f(\xi_p)$ may not hold universally for all distribution families, it is indeed the case for models commonly used in risk modeling practice. For instance, the Pareto distribution considered in our succeeding simulation study in Section 4 satisfies this relationship. This relationship extends, at least at the level of asymptotic behavior, to distributions belonging to the Fréchet domain of attraction, for which the density at ξ_p decays at a polynomial rate with the same functional form as the Pareto case, up to a slowly varying correction. Consequently, heavier tails lead to a larger value of the term $1/f(\xi_p)$, and thus a more pronounced negative bias of the empirical TVaR $\hat{T}_{n,p}$.

In addition, we note that the relationship between the leading bias term ℓ_n and $f(\xi_p)$ stands in stark contrast with that of kernel-based TVaR estimators, whose leading bias term is increasing with $f(\xi_p)$ rather than decreasing (e.g., Chen, 2008; Yu et al., 2010). The reasoning behind this observation may be that a smaller density at ξ_p implies fewer observations and larger spacing between upper order statistics in the neighborhood surrounding ξ_p . As a result, an empirical estimator is less capable of accurately locating the true quantile threshold, which leads to a more pronounced downward bias. However, for kernel methods, a kernel estimator smooths the loss distribution in a neighborhood of the quantile level and therefore assigns positive weight to observations just below ξ_p . These sub-threshold observations are smaller than the true tail losses, and their inclusion pulls the TVaR estimate downward. When $f(\xi_p)$ is larger, a greater amount of probability mass accumulates near ξ_p , which amplifies this smoothing-induced downward pull and can increase the magnitude of the negative bias.

We note that in practical applications, the quantile ξ_p and the PDF f are unknown. To evaluate the leading bias term ℓ_n in (3.6), one may estimate ξ_p using the empirical quantile and estimate $f(\xi_p)$ using a kernel density estimator or another nonparametric method, such as local likelihood density estimation, local polynomial density estimation, or an orthogonal series estimator.

Thus far, Theorem 3.1 identifies the leading term in the asymptotic expansion of the bias B_n of the empirical TVaR estimator. Although informative, the leading bias term ℓ_n need not match the actual bias exactly. To strengthen our analysis of the bias B_n , we next supplement the leading-term approximation in Theorem 3.1 with an explicit upper bound for $|B_n|$, which yields a more conservative evaluation of the bias.

Theorem 3.2. *For a fixed $p \in (0, 1)$, suppose that conditions (C₁) and (C₂) are satisfied. Assume also that the inversion F^{-1} is locally Hölder continuous of order $\gamma \in (0, 1]$ at the point p , that is, there exists a neighborhood $\mathbb{U}_p$ of the point p and a constant $c_\gamma > 0$ such that*

$$|F^{-1}(u) - F^{-1}(p)| \leq c_\gamma |u - p|^\gamma, \quad u \in \mathbb{U}_p. \quad (3.7)$$

Then, for any $\delta > 0$, there exists n_0 such that the bias satisfies

$$-B_n \leq \frac{C p^{\frac{1+\gamma}{2}}}{n^{\frac{1+\gamma}{2}} (1-p)^{\frac{1-\gamma}{2}}}, \quad n \geq n_0, \quad (3.8)$$

where $C = c_\gamma (1 + \delta)$.

Remark 3.6. *The local Hölder continuity constant c_γ in (3.7) quantifies the local smoothness of the quantile function F^{-1} near p . A larger value of c_γ indicates greater steepness or variability of F^{-1} in a neighborhood of p . Hence, a larger value of c_γ implies that even small fluctuations in the sample order statistics may translate into notable errors in the empirical TVaR estimate. Consequently, a larger upper bound is needed to accommodate the increased sensitivity of the negative bias, which explains why the bias bound in (3.8) is monotonically increasing in c_γ when the other parameters are held constant.*

Moreover, since the derivative of the upper bound with respect to c_γ is proportional to $n^{-(1+\gamma)/2}$, its sensitivity to c_γ decreases as the sample size n grows. This implies that for small samples, an accurate assessment of c_γ has a much greater impact on correctly implementing the upper bound than it does for large samples. We will later discuss how to estimate c_γ in practice.

Remark 3.7. *In the choice of $C = c_\gamma(1+\delta)$, the slack parameter $\delta > 0$ is not a structural parameter of the data distribution but an arbitrarily small inflation factor introduced to ensure that the upper bound (3.8) holds uniformly for all sufficiently large n . A smaller δ yields a tighter upper bound in (3.8), but the sample size n may then be larger before the inequality is guaranteed.*

As discussed in Remark 3.6, implementing the negative-bias bound in Theorem 3.2 requires an appropriate evaluation of the local Hölder continuity constant c_γ . First, let us consider an idealized scenario where the underlying data distribution F is known. For almost all commonly

used continuous parametric distributions with a positive and continuous density at ξ_p , the Hölder exponent of F^{-1} near p satisfies $\gamma = 1$. To calculate c_1 , namely the local Hölder constant c_γ when $\gamma = 1$, note that for a small interval $\mathbb{X}_{\xi_p, h} = [\xi_p - h, \xi_p + h]$, $h > 0$, the mean value theorem implies that for all u sufficiently close to p ,

$$|F^{-1}(u) - F^{-1}(p)| \leq \left(\sup_{x \in \mathbb{X}_{\xi_p, h}} \frac{1}{f(x)} \right) |u - p|.$$

Therefore, a local Hölder (or Lipschitz in this case of $\gamma = 1$) constant can be computed via

$$c_1 = \left(\inf_{x \in \mathbb{X}_{\xi_p, h}} f(x) \right)^{-1}. \quad (3.9)$$

In a more realistic scenario where the data distribution is unknown and only data are available, a natural estimator of c_1 is obtained by replacing f with a local density estimator near $\hat{\xi}_p$, the empirical estimate of the p -th quantile. Let $\hat{f}(x)$ denote such an estimator (e.g., kernel density estimator) and let $\mathbb{X}_{\hat{\xi}_p, h} = [\hat{\xi}_p - h, \hat{\xi}_p + h]$ be a small data-driven neighborhood. Then an empirical Lipschitz constant can be computed as

$$\hat{c}_1 = \left(\inf_{x \in \mathbb{X}_{\hat{\xi}_p, h}} \hat{f}(x) \right)^{-1}. \quad (3.10)$$

4 Simulation analysis

The simulation analysis in this section serves two complementary purposes. The first is to assess how well the explicit leading bias term developed in Theorem 3.1 reflects the behavior of the finite-sample bias across varying sample sizes, probability levels, and tail behaviors. The second is to evaluate the performance of the leading-term method from Theorem 3.1 and the bound from Theorem 3.2 in approximating the true bias of empirical TVaR $\hat{T}_{n,p}$. This simulation study does not intend to suggest that the reader should apply the leading bias term and the bias bound as a bias-reduction strategy by default. Instead, this simulation aims to verify that the proposed methods yield a satisfactory assessment of the magnitude of the bias, therefore forming a reliable basis for applying them in the real-data applications that follow. As will be shown in the subsequent real-data illustration section, the bias can be negligible in some scenarios and non-negligible in others, and bias correction should only be considered in the latter.

Throughout this simulation study, we assume that the risk random variable X follows a Pareto

distribution with CDF:

$$F(x) = \begin{cases} 0, & x < 1, \\ 1 - x^{-\alpha}, & x \geq 1, \end{cases} \quad (4.1)$$

where $\alpha > 0$ denotes the shape parameter. The smaller the value of α , the heavier the tail of the distribution of X . The mean of X is finite when $\alpha > 1$, and so is the TVaR T_p for any $p \in (0, 1)$. This is the case of interest throughout this section. In Appendix A, we extend the simulation study to the exponential-tail case and the capped-loss case to demonstrate that the proposed bias analysis framework remains valid across a wide range of data settings.

4.1 Leading-term approximation

Let us begin by examining the leading bias term of the empirical TVaR estimator. Under the model given in (4.1), the mean of the i -th order statistic $X_{i:n}$ based on an iid sample $X_1, \dots, X_n$ from X can be computed explicitly as, for $i < n + 1 - \alpha^{-1}$,

$$\mathbf{E}(X_{i:n}) = \frac{\Gamma(n+1) \Gamma(n-i+1-\alpha^{-1})}{\Gamma(n-i+1) \Gamma(n+1-\alpha^{-1})},$$

where $\Gamma(\cdot)$ represents the gamma function. Consequently, using the representation in the third line of (2.2), the exact bias of the empirical TVaR can be derived explicitly as

$$\begin{aligned} B_n &= \mathbf{E} \left[-\frac{np - [np]}{(1-p)n} X_{[np]+1:n} + \frac{1}{(1-p)n} \sum_{i=[np]+1}^n X_{i:n} \right] - T_p \\ &= \frac{\Gamma(n+1)}{(1-p)n \Gamma(n+1-\alpha^{-1})} \left[-(np - [np]) \frac{\Gamma(n - [np] - \alpha^{-1})}{\Gamma(n - [np])} + \sum_{j=1}^{n-[np]} \frac{\Gamma(j - \alpha^{-1})}{\Gamma(j)} \right] - T_p, \end{aligned}$$

where the true TVaR of (4.1) is given by

$$T_p = \frac{\alpha}{\alpha - 1} (1-p)^{-1/\alpha}, \quad p \in (0, 1).$$

We consider the following baseline parameter setting: $n = 500$, $p = 95\%$, and $\alpha = 3$. We then perturb each parameter individually to examine how the leading bias term ℓ_n varies with n , p , and α , and to verify that the resulting behavior of the bias of the empirical TVaR aligns with the analytical insights established in Section 3. Two versions of the leading bias term ℓ_n are evaluated. The theoretical leading bias term assumes that the data-generating model (4.1) is known, in which case ℓ_n can be computed exactly. The estimated leading bias term considers a

more realistic situation in which only the simulated samples are available. In this case, the kernel density estimator with a Gaussian kernel is used to estimate the unknown PDF appearing in ℓ_n .

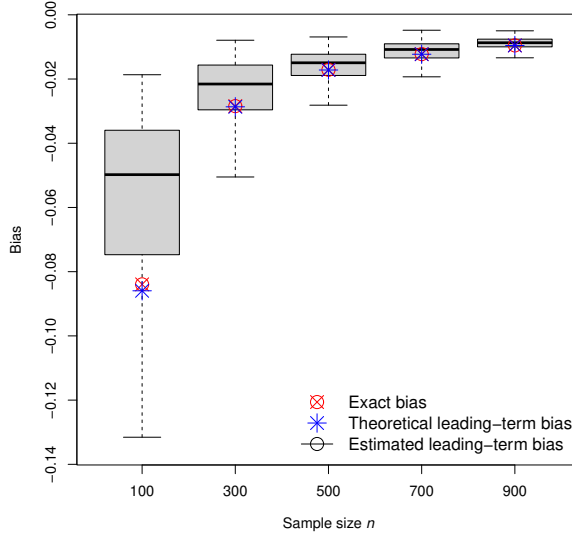


Figure 4.1: Comparison of the exact bias (red circle-cross markers), the theoretical leading-term approximation (blue star markers), and the estimated leading-term approximation (boxplots), with fixed parameters $p = 95\%$, $\alpha = 3$, and varying sample sizes $n \in \{100, 300, 500, 700, 900\}$.

Figure 4.1 presents a comparison of the exact bias with the leading bias term ℓ_n derived in Theorem 3.1 with varying sample sizes. In the figure, the boxplots are constructed from $m = 100$ simulated datasets of size n in order to account for simulation variation. As shown, when the data-generating distribution is assumed to be known, the exact bias and its leading-term approximation agree very closely. Among the sample sizes considered, a visible yet minor discrepancy is observed only when $n = 100$. This indicates that, at least for the Pareto model, the residual term in the asymptotic expansion (3.6) is negligible even at small sample sizes. When the distribution is assumed to be unknown and $f(\xi_p)$ must be estimated from data, the performance of the leading-term approximation deteriorates due to the limited accuracy of the kernel-based estimation of $f(\xi_p)$. For smaller samples (e.g., $n = 100$ or 300), the true bias falls near or just outside the whiskers of the boxplots, though even in these cases the estimates remain reasonably close to the true value and still provide a meaningful approximation of the bias. However, as the sample size increases, the estimation of $f(\xi_p)$ becomes more accurate and stable. For larger sample sizes (e.g., $n = 900$), the leading-term approximation based on the kernel density estimate tracks the exact bias of the empirical TVaR very closely.

Next, we examine the impact of the probability level p on both the exact bias B_n and the leading bias term ℓ_n . As p increases, fewer observations fall beyond the quantile threshold ξ_p , thus reducing

the amount of data available for estimating TVaR. Consequently, the negative bias of the empirical TVaR estimator becomes more pronounced. This pattern observed in Figure 4.2 is consistent with the analytical insights discussed in Remark 3.4. When the data-generating distribution is treated as unknown, increasing p also deteriorates the accuracy of the kernel density estimator for $f(\xi_p)$, leading to a larger discrepancy between the estimated leading bias term and the exact bias. Nevertheless, the theoretical leading bias term tracks the exact bias very closely across all probability levels considered. This suggests that the discrepancy observed in the estimated leading bias term is mainly attributable to the estimation error in the kernel density estimator, rather than to the truncation of the asymptotic expansion in (3.6).

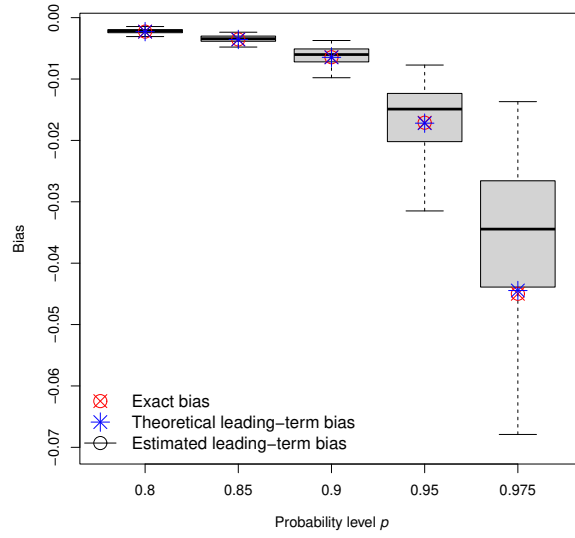


Figure 4.2: Comparison of the exact bias (red circle-cross markers), the theoretical leading-term approximation (blue star markers), and the estimated leading-term approximation (boxplots), with fixed parameters $n = 500$, $\alpha = 3$, and varying probability levels $p \in \{80, 85, 90, 95, 97.5\}\%$.

Further, we study the impact of the tail heaviness of the data distribution specified in (4.1) on the bias of the empirical TVaR B_n and its leading-term approximation ℓ_n . The tail weight of the Pareto model (4.1) is controlled by the shape parameter α . The results are summarized in Figure 4.3. As shown, smaller values of α , corresponding to heavier tails, lead to a larger magnitude of the negative bias, for both the exact bias and its leading-term approximation. This observation is consistent with the analytical insights discussed in Remark 3.5. In particular, under the data model in (4.1), elementary algebraic calculations yield

$$f(\xi_p) = \alpha (1 - p)^{(\alpha+1)/\alpha},$$

which is increasing in $\alpha > 0$ for a fixed $p \in (0, 1)$. Thus, smaller values of α , i.e., heavier tails, lead to smaller values of $f(\xi_p)$, which in turn amplify the negative leading-term bias. This behavior of ℓ_n matches that of the exact bias B_n across all values of α considered. We also observe from Figure 4.3 that heavier tails impair the accuracy of estimating $f(\xi_p)$ via kernel density estimation, thus resulting in a more noticeable discrepancy between the estimated leading bias term and the exact bias when α is small.

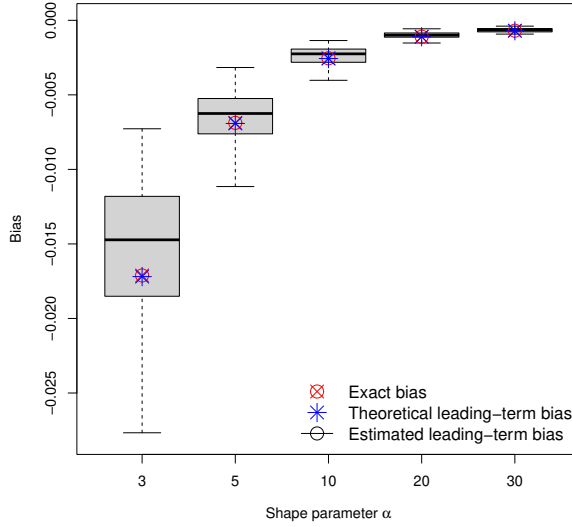


Figure 4.3: Comparison of the exact bias (red circle-cross markers), the theoretical leading-term approximation (blue star markers), and the estimated leading-term approximation (box-plots), with fixed parameters $n = 500$, $p = 95\%$, and varying values of the shape parameter $\alpha \in \{3, 5, 10, 20, 30\}$.

Finally, we remark that in the high-probability regime in Figure 4.2, the leading-term approximation, as an estimate of the bias of the empirical TVaR, is itself biased. This secondary bias is caused by the kernel density estimation involved in evaluating the leading term ℓ_n . It is known that the kernel density estimator is biased, and the direction of the bias is determined by the sign of the second derivative of the underlying density function (e.g., [Silverman, 1998](#)). Because the density of the Pareto distribution assumed in (4.1) is convex, the bias of the kernel estimator is positive. Consequently, the kernel density estimator tends to overestimate $f(\xi_p)$, resulting in an underestimation of $|\ell_n|$. To mitigate this issue, one may use higher-order kernels or local polynomial methods to reduce the bias in density estimation, typically at the expense of increased variance. Meanwhile, it is noteworthy that perfectly eliminating the bias of the empirical TVaR estimator in finite-sample applications is unrealistic. When a significant bias is detected and bias reduction is warranted, the estimated leading term ℓ_n should be used as a means of mitigating the bias rather

than eliminating it entirely.

4.2 Negative-bias upper bound

We now turn to examining how well the negative-bias upper bound in Theorem 3.2 captures the true bias. To implement the bound, it is necessary to specify two tuning parameters. The first one is the slack parameter δ , which controls the trade-off between the tightness of the bound and the sample size required for its validity in finite samples (see Remark 3.7 for a detailed discussion). The second one is the locality parameter h , which determines the local Lipschitz constant through (3.9). Although introduced from different perspectives, note that these two tuning parameters in fact play similar roles in practice. Specifically, a larger value of the locality parameter h enlarges the interval over which the Lipschitz constant is evaluated, resulting in a larger estimate of c_1 and, consequently, a more conservative bound. This effect parallels the role of the slack parameter δ discussed in Remark 3.7. For this reason, throughout the analysis in this subsection, we fix the locality parameter $h = 0.05$ and focus on examining the sensitivity of the bias bound with respect to different choices of the slack parameter δ .

Suppose that the data-generating model (4.1) is known. For a given locality parameter h , the local Lipschitz constant defined in (3.9) admits the following closed-form expression:

$$c_1 = \left(\inf_{x \in \mathbb{X}_{\xi_p, h}} f(x) \right)^{-1} = \frac{(\xi_p + h)^{\alpha+1}}{\alpha} = \frac{((1-p)^{-1/\alpha} + h)^{\alpha+1}}{\alpha},$$

where the second equality follows from the fact that the PDF of the Pareto distribution in (4.1) is strictly decreasing. Now, we vary the sample size n to examine whether the negative-bias upper bound remains valid for small sample sizes and, if so, how tight the bound is across different values of n . The comparison results are summarized in Figure 4.4. As can be observed in Figure 4.4, the bound always covers the true bias across all values of the slack parameter δ and sample sizes n considered. For a fixed δ , the bound captures the decreasing trend of the exact negative bias as the sample size n increases. For a fixed n , larger values of the slack parameter δ lead to a more conservative bound and thus a larger discrepancy from the exact bias, which is consistent with the discussion in Remark 3.7. Interestingly, the gap between the exact bias and the bound is substantially larger for smaller sample sizes and lessens noticeably as n increases. This observation can be attributed to the presence of positive residual terms in the derived upper bound under the Pareto model (4.1), which reduce as the sample size grows. As a result, the bound becomes increasingly tight in larger samples.

In another scenario where the data-generating model (4.1) is treated as unknown, the kernel-based estimator (3.10) is employed to estimate the Lipschitz constant required for computing the

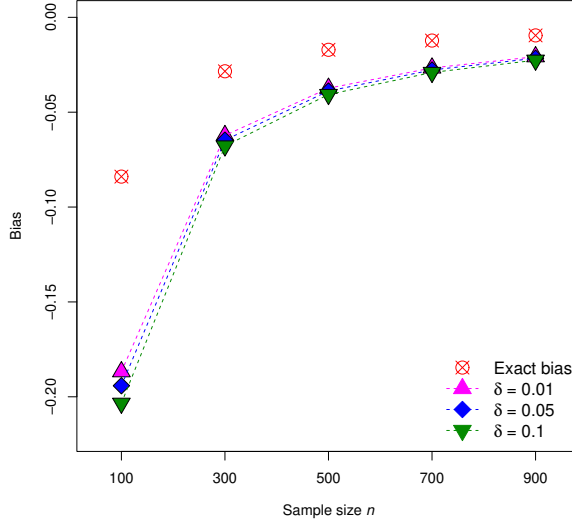


Figure 4.4: Comparison of the exact bias with the theoretical negative-bias upper bound, with fixed parameters $p = 95\%$, $\alpha = 3$, and $h = 0.05$, across varying sample sizes $n \in \{100, 300, 500, 700, 900\}$ and different choices of the slack parameter $\delta \in \{0.01, 0.05, 0.1\}$.

bias bound. Similar to the numerical experiments conducted in the previous subsection, each configuration is repeated $m = 100$ times to account for the sampling variability inherent in the simulated data. The resulting performance of the bound relative to the exact negative bias is summarized in Figure 4.5. Several observations can be made. When the sample size is small, the kernel density estimator more frequently misestimates the PDF in the tail region, leading to a noticeable discrepancy between the center of the boxplots and the theoretical bias bound. As a result, the estimated bias bound may occasionally fail to capture the exact bias. As the sample size increases, the kernel density estimator becomes more accurate and stable. Consequently, the boxplots become narrower, and their centers converge toward the theoretical bias bound. In this example, we observe that once the sample size exceeds 500, the estimated bias bound consistently covers the true bias, which indicates improved finite-sample reliability of the negative-bias upper bound as the sample size grows.

5 Real-data illustration

This section aims to demonstrate the practical utility of the bias analysis framework established in Section 3 for studying the bias severity question Q1. The first real-data example represents a regime in which the sample size is at least moderate. In this regime, the framework suggests that the bias is small and perhaps negligible. However, in the second example, where data are limited

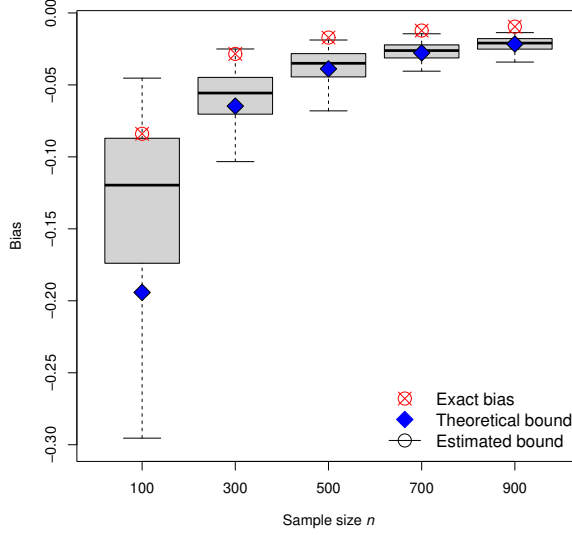


Figure 4.5: Comparison of the exact bias (red circle-cross markers), the theoretical negative-bias upper bound (blue diamond markers), and the estimated negative-bias upper bound (boxplots), with fixed parameters $p = 95\%$, $\alpha = 3$, $\delta = 0.05$, and $h = 0.05$, across varying sample sizes $n \in \{100, 300, 500, 700, 900\}$.

and exhibit heavy tails, a contrasting, cautionary regime emerges, and the bias can represent a substantial fraction of the value of TVaR.

5.1 An example with sufficient sample size

Our first real-data example is based on the Danish fire loss dataset (McNeil, 1997). The data consist of 2167 individual fire insurance losses recorded between 1980 and 1990, which have been inflation-adjusted to 1985 price levels and are reported in millions of Danish kroner. Prior studies of the dataset have documented that its distribution exhibits a heavy right tail, with a tail index of approximately 1.4 (e.g., McNeil, 1997; Resnick, 1997). As demonstrated by the simulation experiments in Section 4, heavy-tailed data can induce an increased bias in $\hat{T}_{n,p}$, particularly at high probability levels. This motivates us to examine the magnitude of the bias of the estimator.

Although the tail of the distribution of the Danish fire data is relatively heavy, as indicated by the estimated tail index of approximately 1.4, the distribution still has a finite first moment. Consequently, condition (C₁) is satisfied. Moreover, since the loss data are strictly positive, condition (C₂) also holds. These conditions together justify the validity of using the leading-term bias approximation in Theorem 3.1, as well as the bias bound in Theorem 3.2, to analyze the bias of the empirical TVaR estimator in this data setting.

In the absence of knowledge of the true underlying loss distribution, unlike in the simulation

study, the exact bias is not available for direct comparison. To address this, we compare the proposed leading bias term and bias bound against estimates obtained using a classical numerical approach, such as the bootstrap method. Specifically, we employ the nonparametric Efron bootstrap with $l = 1\,000$ resample sets, each of size $n = 2\,167$, which is the same as the size of the original dataset. To account for the variability inherent in the bootstrap procedure, we repeat the bootstrap bias estimation $m = 100$ times.

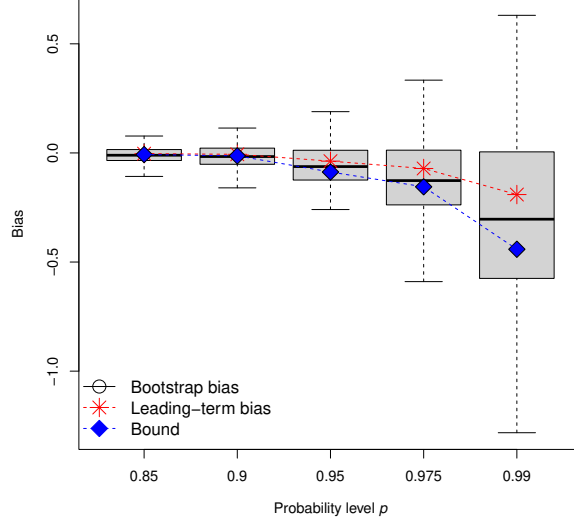


Figure 5.1: Comparison of bias estimates for the empirical TVaR based on the Danish fire loss dataset, obtained via the leading-term approximation (red star markers), the negative-bias upper bound (blue diamond markers), and the bootstrap method (boxplots).

Figure 5.1 depicts a comparison of bias estimates for the empirical TVaR obtained using different methods, including the leading-term approximation and the negative-bias upper bound derived in Theorems 3.1 and 3.2, respectively, as well as the bootstrap method. We observe that, due to resampling variability, the bootstrap procedure may occasionally produce positive bias estimates. Such estimates contradict the theoretical result that the bias of empirical TVaR B_n is always negative (Wei and Zitikis, 2023; Gribkova et al., 2026); also see discussion in Remark 3.1. For the two methods proposed in this paper, owing to their analytical nature, they do not suffer from this drawback induced by resampling variation, and therefore provide bias assessments that are always consistent with the negativity property.

At lower probability levels, the leading-term approximation matches closely with the means of the bootstrap estimates. At higher probability levels, the leading-term approximation exhibits more pronounced discrepancies from the mean of the bootstrap estimates. At the same time, the bootstrap estimates themselves display increased variability, which is caused by the greater sampling

uncertainty associated with observations in the extreme tail region that drive the estimation of TVaR. Moreover, across all probability levels considered, the bias bound consistently covers both the leading-term approximation and the means of the bootstrap bias estimates, while remaining reasonably close to the lower edge of the bootstrap boxplot whiskers. This behavior indicates a satisfactory performance of the bias bound in terms of achieving a practical balance between tightness and conservativeness.

The bias analysis framework established in this paper provides practitioners with a deterministic, efficient, and transparent method for assessing the significance of bias, thus supporting informed decision-making regarding whether bias-reduction strategies are warranted or whether additional data collection is desirable. In this data example, $\hat{T}_{n,p}$ yields a TVaR estimate of approximately 24 at the 95% probability level, while the corresponding bias of the empirical TVaR is estimated to be around -0.1 , representing less than 0.5% of the TVaR value. This is an affirmative finding, as such a negligible level of bias indicates that the empirical TVaR estimator can be used with confidence even without applying any bias correction.

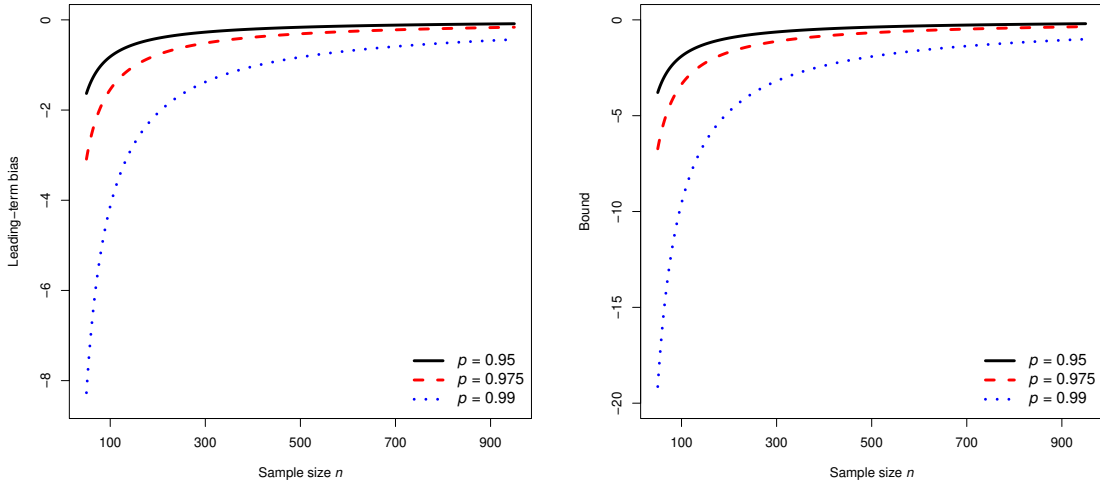


Figure 5.2: Leading bias term (left panel) and bias bound (right panel) of the empirical TVaR as functions of the sample size n , evaluated at probability levels $p \in \{95, 97.5, 99\}\%$.

Moreover, the derived leading bias term and the bias bound are useful for analysts seeking to understand how the choice of sample size and probability level affects the reliability of TVaR estimation. In Figure 5.2, we first use the original Danish fire loss data to estimate $f(\xi_p)$ and the Lipschitz constant c_1 , which are the inputs for calculating the leading bias term ℓ_n in (3.6) and the bias bound in (3.8). We then plot the resulting bias against sample size across different probability levels. We observe that the growth of the bias intensifies sharply once the sample size falls below approximately 150 for $p = 95\%$, 200 for $p = 97.5\%$, and 300 for $p = 99\%$. When

obtaining additional data is possible, these relationships indicate the sample sizes at which the marginal benefit of acquiring extra data is greatest. When additional data are not available, they instead offer insight into the small-sample regime in which the bias becomes severe, which motivates the example that follows.

5.2 An example with constrained sample size

The second example is based on the Asia-Pacific (APAC) large commercial risks (LCR) dataset, assembled by [Benedetti et al. \(2015\)](#). The dataset contains anonymized LCR claims caused by man-made risks such as fire and explosion, related to commercial lines of business including commercial, manufacturing, and energy exposures.

The dataset can be accessed via the R package “CASdatasets”.¹ The losses contained in the dataset are expressed in millions of euros. They span the years 2000 to 2013, and are grouped into four time periods, one well before the 2008 financial crisis, one immediately before the crisis, one during the crisis, and one after the crisis. It is worth stressing that these periods are not defined artificially by this paper. Rather, they reflect how the original data are organized when they are extracted from the R package. Table 5.1 summarizes the key distributional features of the data across the four periods. Structural differences across periods are evident. First, while the number of recorded claims in each period ranges similarly between 100 and 200, the claim frequency, viewed as the number of claims relative to the length of the period, differs noticeably, with the 2009–2010 crisis period exhibiting the highest frequency. Second, over time, the percentage of claims originating from companies operating in developed countries appears to decrease. Among the four periods, the first two exhibit higher loss amounts than the last two, in terms of both their quantiles and means. This observed difference over time should not be interpreted solely as reflecting improved risk management practices among companies in developed countries. In addition, there has been a change in the data collection process. As stated in the CASdatasets documentation, the data initially focused on claims exceeding 1 million in losses, but were later extended to include claims exceeding 0.14 million. Moreover, the loss data in all four periods exhibit a heavy-tailed pattern, with tail index estimates ranging between 1 and 2. Comparing the periods, the 2009–2010 loss data exhibit the heaviest tail, though we acknowledge that, given the size of the available data, the confidence intervals of the tail indexes across periods may overlap.

We suppose that an insurer providing coverage for individual LCR claims of the kind captured in this dataset, is interested in applying the TVaR risk measure at $p = 97.5\%$ to assess the riskiness of the claim severity, and how this riskiness evolves over time. We further suppose the empirical TVaR estimator is used to estimate the TVaR for each period separately. The structural heterogeneity

¹Source: <https://cas.uqam.ca/>

Period	n	Dev. %	$\xi_{0.25}$	$\xi_{0.5}$	$\xi_{0.75}$	Average	Tail index estimate
2000–2003	111	67.6	1.93	8.81	22.58	23.52	1.95
2004–2008	176	42.0	0.53	3.85	20.17	20.92	1.51
2009–2010	116	17.2	0.30	0.83	6.66	7.98	1.35
2011–2013	123	17.9	0.41	1.00	6.04	7.36	1.55

Table 5.1: Summary of key distributional properties of the losses in the APAC LCR dataset. In the table, “Dev. %” represents the percentage of claims originating from developed countries as defined by the World Bank’s economic development classification; $\xi_{0.25}$, $\xi_{0.5}$, $\xi_{0.75}$ are the 25th, 50th, 75th percentiles of the losses, respectively; the tail index is estimated based on the top 10% of order statistics over each period using the Hill estimator.

discussed above justifies analyzing the loss data period by period, which ensures a reasonable degree of homogeneity within each period and thereby supports the iid assumption required by the bias analysis framework.

We apply the leading-term method and the bias bound to analyze the bias of the empirical TVaR estimator. To ensure that the two proposed methods perform well in this data setting, as in the previous example in Section 5.1, we first compare both to a numerical bootstrap method. The comparison is shown in Figure 5.3. As can be observed, the leading bias term tracks the bootstrap interval estimates closely across all periods, and the bias bound covers the bootstrap estimates in every period considered. It is worth noting that the bootstrap estimation intervals in Figure 5.3 are relatively narrow compared to the bias bounds. This is because we adopt $l = 1000$ resample sets, each equal in size to the sample of the corresponding period, to obtain more stable bootstrap bias estimates. It is important to note that these intervals reflect the variability of the bootstrap-based bias estimator over $l = 1000$ resample sets, not the variability of the empirical TVaR estimator itself.

After the reliability of the leading-term method and the bias bound has been confirmed in this data setting, Table 5.2 summarizes the severity of the estimated bias of the empirical TVaR estimator and its magnitude relative to the TVaR itself. Both the leading-term and bound-based bias estimates are most severe relative to the TVaR level when the empirical TVaR is applied to the 2009–2010 period data. The severity of the bias is comparable between the 2004–2008 and 2011–2013 periods, and is lowest for the 2000–2003 period. This ordering coincides with the ordering of the tail weights implied by the estimated tail indices in Table 5.1, consistent with the analytical insights predicted by Theorem 3.1 (also see Remark 3.5).

Focusing on the results of the leading-term method, the bias of the empirical TVaR is more than 12% of the estimated value of TVaR in the 2009–2010 period. The significance of the bias relative to the estimator’s variability is also studied. Recall that the estimated tail indices of the

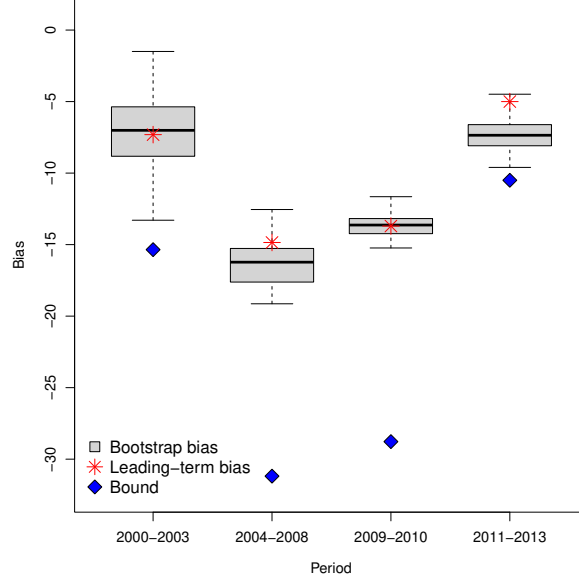


Figure 5.3: Comparison of bias estimates for the empirical TVaR based on the APAC LCR dataset, obtained via the leading-term approximation (red star markers), the negative-bias upper bound (blue diamond markers), and the bootstrap method (boxplots).

Period	$\hat{T}_{n,p}$	$\text{MAD}(\hat{T}_{n,p}^*)$	Bias leading term	Bias bound
2000–2003	210.23	64.31	−7.31 (−3.48%)	−15.35 (−7.30%)
2004–2008	259.85	42.19	−14.85 (−5.72%)	−31.19 (−12.00%)
2009–2010	112.22	19.35	−13.70 (−12.21%)	−28.77 (−25.64%)
2011–2013	108.08	31.87	−5.00 (−4.63%)	−10.50 (−9.72%)

Table 5.2: Summary of the values of empirical TVaR at $p = 97.5\%$, the mean absolute deviations (MAD) of the empirical TVaR estimator, the bias leading terms derived in Theorem 3.1, and the bias bounds derived in Theorem 3.2. The values in parentheses are the percentages relative to the estimated TVaR $\hat{T}_{n,p}$.

loss data across the four periods range between 1 and 2 (see Table 5.1). Because of this heavy-tailed feature, the variance of the empirical TVaR estimator does not exist. We therefore use the mean absolute deviation (MAD) as a robust alternative which remains finite given the estimated tail indices. Comparing the MAD with the magnitude of the leading bias term shows that, in the most severe period (2009–2010), the bias is about 70% of the MAD of the empirical TVaR estimator. Together, this suggests that the bias is not negligible. Caution and bias reduction are warranted in this data environment. In this respect, the leading bias term calculated can be readily used to reduce the bias.

Regarding the bias bounds, they estimate the bias to be as high as 25% of the estimated

value of TVaR in the most severe period, 2009–2010. Although these estimates are likely overly conservative, they can be useful in practice as a safety buffer against underestimation of riskiness or undercapitalization when TVaR is used to determine capital requirements. In this respect, we note that regulators explicitly advocate adding a margin of conservatism to guard against over-optimism in risk estimates, and the margin should be larger when the underlying data are less satisfactory (see Paragraph 451 in [BCBS, 2006](#)). The bias bounds derived here share the same spirit as this regulator-advocated margin of conservatism, and thus offer a practical approach for determining an appropriate size of this conservatism margin.

6 Conclusions

TVaR has been widely adopted by practitioners and regulators as a conservative risk measure for capturing tail risk. In practical applications, TVaR is typically estimated using the empirical method due to its nonparametric nature and ease of implementation. It is known that the empirical TVaR estimator exhibits a negative finite-sample bias ([Wei and Zitikis, 2023](#); [Gribkova et al., 2026](#)). However, despite its practical importance, the magnitude of this bias and its dependence on the distributional characteristics of the data and the sample size remain underexplored.

This paper addresses this gap and thus answers question [Q1](#) posed at the beginning of this article, by developing two analytical approaches for evaluating the bias of the empirical TVaR. The first approach is based on the leading term in the asymptotic expansion of the bias, while the second approach establishes an explicit bound on the bias. These results yield explicit analytical insights into how data distributional properties, sample sizes, and probability levels jointly determine the magnitude of the bias. Simulation studies demonstrate that the proposed methods perform satisfactorily in approximating the exact bias across a wide range of data settings. We further apply the proposed methods to study the bias involved in TVaR estimation using two real-world datasets, and demonstrate the two-sided answer to question [Q1](#).

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Appendix A Additional numerical experiments

A.1 Robustness check against light-tailed distributions

The simulation study in Section 4 focuses on the Pareto distribution, whose right tail exhibits a power form. In this part of the appendix, we supplement the study with another model whose tail decays exponentially and is thus lighter than that of the Pareto distribution. The goal of this additional simulation is to demonstrate that the performance of the proposed leading-term and bound-based methods remains satisfactory for estimating the bias of the empirical TVaR in the case of exponential tails, and that the empirical TVaR performs more reliably in this setting than in the Pareto case.

For $\theta > 0$, we assume that a risk position is governed by the shifted exponential distribution with CDF:

$$F(x) = \begin{cases} 0, & x < 1, \\ 1 - \exp\{- (x - 1)\theta\}, & x \geq 1. \end{cases} \quad (\text{A.1})$$

The shifted exponential distribution in (A.1) has the same support as the Pareto distribution in (4.1), which ensures a consistent comparison of the reliability of the empirical TVaR between the two settings. For each $p \in (0, 1)$ considered, we calibrate θ such that the p -th quantile ξ_p is the same as the one implied by the baseline Pareto setting with $\alpha = 3$.

Figure A.1 compares the bias analysis based on the leading-term method between the shifted exponential case and the Pareto case, across different sample sizes n and probability levels p . As shown, the leading-term approximation tracks the true bias closely when the model is known, and, when the model is unknown, the estimated leading term still reasonably captures the true bias. The sensitivity pattern of the bias in response to changing n and p in the exponential case is consistent with what the analytic leading term in Theorem 3.1 predicts, and is largely the same as the pattern exhibited in the Pareto case.

Figure A.2 displays the same comparison for the bias bound. As with the leading-term method, the bias bounds, including both the theoretical ones (assuming the data model is known) and the estimated ones (assuming the data model is unknown), are capable of covering the true bias.

Moreover, it can be seen from Figures A.1 and A.2 that the magnitude of the bias of the empirical TVaR is smaller in the exponential case than in the Pareto case. At the same time, when

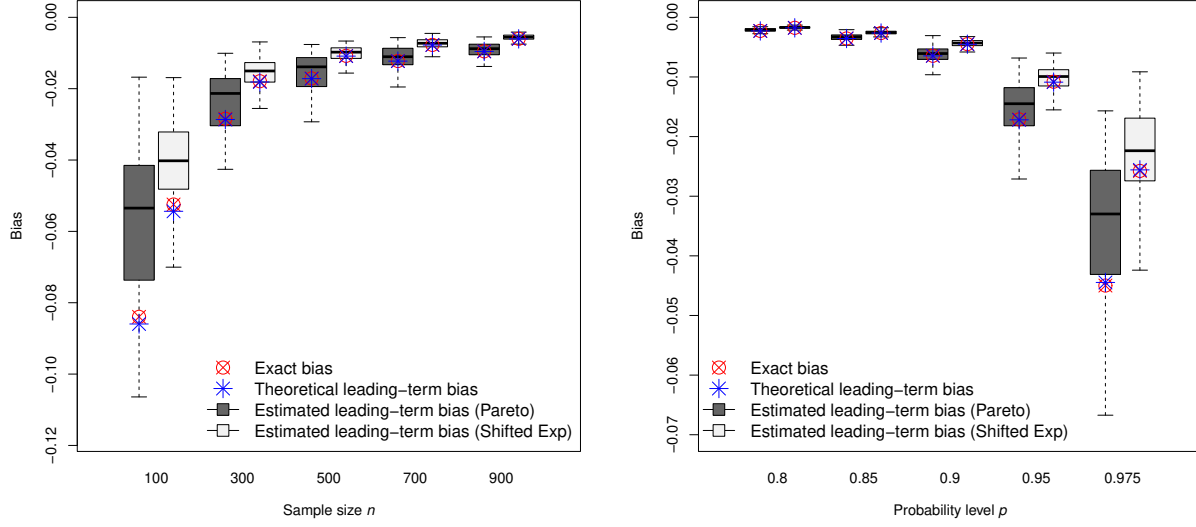


Figure A.1: Comparison of the bias leading terms between the exponential model (A.1) and the Pareto model (4.1). The left panel fixes $p = 95\%$ and varies n ; the right panel fixes $n = 500$ and varies p .

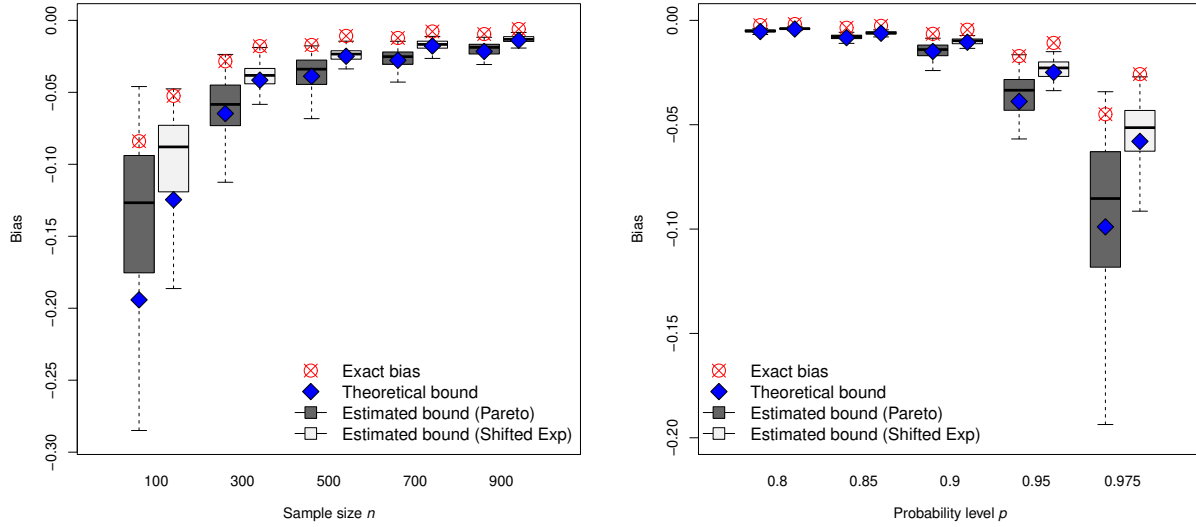


Figure A.2: Comparison of the bias bounds between the exponential model (A.1) and the Pareto model in (4.1). The left panel fixes $p = 95\%$ and varies n ; the right panel fixes $n = 500$ and varies p .

the model is unknown, both the leading bias term and the bias bound in the exponential case are more convenient to estimate, in the sense that the associated estimates exhibit noticeably smaller variance. Collectively, these findings allow us to conclude that the empirical TVaR performs more

reliably in the exponential case than in the Pareto case, which is in agreement with the theoretical finding established in Remark 3.5 that heavy-tailed data result in a more severe bias.

A.2 Robustness check in the case of capped losses

In the introduction, we argued that one advantage of pursuing the more technically demanding approach taken in this paper is the minimal assumptions it requires, which allow our results to extend to data settings in which the loss is subject to an upper limit, for example due to a policy limit or a reinsurance treaty. In this numerical experiment, we retain the baseline Pareto setting from Section 4, with $\alpha = 3$ and $p = 95\%$, but apply an upper limit at ξ_{p^*} , with $p^* = 98\%$, such that the observed data are represented by $X^* = \min(X, \xi_{p^*})$, where X follows the Pareto distribution in (4.1) and ξ_{p^*} is the p^* -th quantile of X .

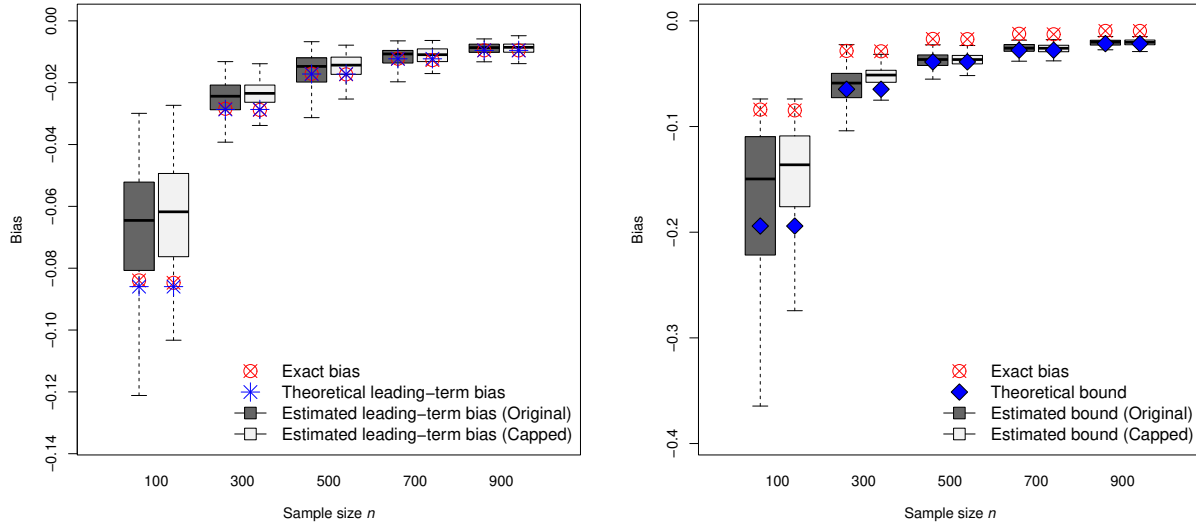


Figure A.3: Comparison of the leading bias term (left panel) and the bias bound (right panel) between data with and without an upper limit, simulated based on the Pareto model in (4.1).

The performance of the leading bias term and the bias bound across different sample sizes n is summarized in Figure A.3. As shown, both methods perform well in capturing the true bias when the data are subject to this upper limit. We also observe that the actual bias of the empirical TVaR is highly similar, regardless of whether or not the upper limit is applied. This is consistent with Theorems 3.1 and 3.2, whose expressions depend only on the density behavior around ξ_p , and not on the tail behavior beyond that point, which is the region affected by the upper limit.

We stress that the standard approach discussed in the introduction, which relies on the asymptotic expansion of the empirical quantile function and thus requires the differentiability of the

density throughout the entire tail, does not extend to this setting. Consequently, under the standard approach, the resulting approximation would not be valid in this capped data setting, and the corresponding analytical insight could not be obtained. This underscores the practical value of the more demanding route toward the theoretical results developed in this paper.

Appendix B Technical proofs

Let us introduce some additional notation to facilitate the succeeding technical developments. Recall that the winsorized version of X_i with respect to the threshold ξ_p is denoted by $W_i = \max(X_i, \xi_p)$, $i = 1, \dots, n$. The random variables $W_1, \dots, W_n$ are iid. The common CDF of $W_1, \dots, W_n$ can be computed via

$$F_W(x) = \begin{cases} 0, & x < \xi_p, \\ F(x), & x \geq \xi_p, \end{cases}$$

and the corresponding quantile function is given by

$$F_W^{-1}(u) = \max(\xi_p, F^{-1}(u)) = \begin{cases} \xi_p, & u \leq p, \\ F^{-1}(u), & u > p. \end{cases}$$

The common mean of $W_1, \dots, W_n$, shorthanded by $\mu_W = \mathbf{E}(W_1)$, can be expressed as

$$\mu_W = \int_0^1 F_W^{-1}(u) \, du = p \xi_p + \int_p^1 F^{-1}(u) \, du. \quad (\text{B.1})$$

Moreover, let $W_{1:n} \leq \dots \leq W_{n:n}$ denote the order statistics based on the random variables $W_1, \dots, W_n$. For $i = 1, \dots, n$, it holds that

$$W_{i:n} = \begin{cases} \xi_p, & i \leq N_p, \\ X_{i:n}, & i > N_p. \end{cases}$$

Proof of Lemma 3.1. Recall the representation given in the third line of (2.2). Multiplying both sides by $(1 - p)$, we get:

$$(1 - p) \hat{T}_{n,p} = -\frac{np - [np]}{n} X_{[np]+1:n} + \frac{1}{n} \sum_{i=[np]+1}^n X_{i:n}.$$

Consider the average of $W_1, \dots, W_n$. We have

$$\frac{1}{n} \sum_{i=1}^n W_i = \frac{1}{n} \sum_{i=1}^n W_{i:n} = \frac{N_p}{n} \xi_p + \frac{1}{n} \sum_{i=N_p+1}^n X_{i:n},$$

because $W_{i:n} = X_{i:n}$ for all $i > N_p$. Taking into account (B.1), we find

$$\begin{aligned} (1-p)(\hat{T}_{n,p} - T_p) - \frac{1}{n} \sum_{i=1}^n (W_i - \mu_W) &= -\frac{np - [np]}{n} X_{[np]+1:n} + \frac{1}{n} \sum_{i=[np]+1}^n X_{i:n} - \frac{N_p}{n} \xi_p - \frac{1}{n} \sum_{i=N_p+1}^n X_{i:n} \\ &\quad - \underbrace{\int_p^1 F^{-1}(u) du + p \xi_p + \int_p^1 F^{-1}(u) du}_{\mu_W} \\ &= -\frac{np - [np]}{n} (X_{[np]+1:n} - \xi_p) - \frac{N_p - [np]}{n} \xi_p + S_n, \end{aligned} \quad (\text{B.2})$$

where

$$S_n = \frac{1}{n} \sum_{i=[np]+1}^n X_{i:n} - \frac{1}{n} \sum_{i=N_p+1}^n X_{i:n} = \frac{\text{sign}(N_p - [np])}{n} \sum_{i=(\lfloor np \rfloor \wedge N_p)+1}^{\lfloor np \rfloor \vee N_p} X_{i:n}. \quad (\text{B.3})$$

Combining (B.2) and (B.3), we obtain

$$\begin{aligned} \hat{T}_{n,p} - T_p - \frac{1}{n(1-p)} \sum_{i=1}^n (W_i - \mu_W) &= -\frac{np - [np]}{n(1-p)} (X_{[np]+1:n} - \xi_p) + \frac{\text{sign}(N_p - [np])}{n(1-p)} \sum_{i=(\lfloor np \rfloor \wedge N_p)+1}^{\lfloor np \rfloor \vee N_p} (X_{i:n} - \xi_p) \\ &= -R_n, \end{aligned}$$

which implies (3.2) and (3.3).

Next, we prove (3.4). There are three cases: (i) $[np] = N_p$; (ii) $[np] < N_p$; and (iii) $[np] > N_p$. For case (i), we have $X_{[np]+1:n} - \xi_p > 0$, hence the first term of R_n is not negative, whereas its second term is zero because, according to the definition of the sign function in (3.1), $\text{sign}(N_p - [np]) = 0$.

For case (ii), we have $X_{i:n} \leq \xi_p$ for $i = [np] + 1, \dots, N_p$, and hence,

$$\begin{aligned} R_n &= \frac{np - [np]}{n(1-p)} (X_{[np]+1:n} - \xi_p) - \frac{1}{n(1-p)} \sum_{i=[np]+1}^{N_p} (X_{i:n} - \xi_p) \\ &= \frac{np - [np] - 1}{n(1-p)} (X_{[np]+1:n} - \xi_p) - \frac{\mathbb{1}_{\{N_p > [np]+1\}}}{n(1-p)} \sum_{i=[np]+2}^{N_p} (X_{i:n} - \xi_p) \geq 0. \end{aligned}$$

Here, it is noteworthy that the indicator $\mathbb{1}_{\{N_p > [np] + 1\}}$ in front of the second sum is needed because if $N_p = [np] + 1$, then the second sum disappears.

For case (iii), we have $\text{sign}(N_p - [np]) = -1$, and $X_{i:n} > \xi_p$ for all $i > N_p$. Hence,

$$R_n = \frac{np - [np]}{n(1-p)} (X_{[np]+1:n} - \xi_p) + \frac{1}{n(1-p)} \sum_{i=N_p+1}^{[np]} (X_{i:n} - \xi_p) \geq 0.$$

The proof for this assertion is now finished. $\square$

To establish the main theorems in Section 3, the following assertion serves as an important auxiliary result. Specifically, let k and ε be arbitrary positive numbers, and set $\rho := k/\varepsilon$. The following lemma suggests an upper bound estimate for the absolute moment of the i -th order statistic, $\mathbf{E}|X_{i:n}|^k$, for $\rho \leq i \leq n - \rho + 1$, in terms of the absolute moment of the underlying distribution, $\mathbf{E}|X|^\varepsilon$. The proof of the lemma can be found in [Gribkova \(1995\)](#).

Lemma B.1 ([Gribkova, 1995](#)). *For all $n \geq 2\rho + 1$ and all i such that $\rho \leq i \leq n - \rho + 1$, the following inequality holds:*

$$\mathbf{E}|X_{i:n}|^k \leq C(\rho) \left\{ \mathbf{E}|X|^\varepsilon h\left(\frac{i}{n+1}\right) \right\}^\rho, \quad (\text{B.4})$$

where the function $h : (0, 1) \rightarrow (0, \infty)$ is given by $h(u) = [u(1-u)]^{-1}$, $u \in (0, 1)$, and the constant $C(\rho)$ may be taken as $C(\rho) = 2\sqrt{\rho} \exp(\rho + 7/6)$.

To facilitate the presentation of the proofs of Theorems 3.1 and 3.2, some additional notations are needed. Let $U_1, \dots, U_n$ be independent $[0, 1]$ -uniform random variables, and $U_{1:n} \leq \dots \leq U_{n:n}$ represent their order statistics. For a fixed $p \in (0, 1)$, define the binomial random variable $M_p = \text{card}\{i : U_i \leq p\}$, where $\text{card}\{\cdot\}$ denotes the cardinality of a set. Then, since the joint distribution of the order statistics $X_{1:n}, \dots, X_{n:n}$ is identical to that of $F^{-1}(U_{1:n}), \dots, F^{-1}(U_{n:n})$, the remainder term R_n in (3.3) can be written as

$$R_n \stackrel{d}{=} R_{n,1} + R_{n,2}, \quad (\text{B.5})$$

where

$$\begin{aligned} R_{n,1} &:= \frac{np - [np]}{n(1-p)} (F^{-1}(U_{[np]+1:n}) - F^{-1}(p)), \\ R_{n,2} &:= - \frac{\text{sign}(M_p - [np])}{n(1-p)} \sum_{i=([np] \wedge M_p) + 1}^{[np] \vee M_p} (F^{-1}(U_{i:n}) - F^{-1}(p)). \end{aligned} \quad (\text{B.6})$$

Accordingly, we have

$$-B_n = \mathbf{E}(R_n) = \mathbf{E}(R_{n,1}) + \mathbf{E}(R_{n,2}). \quad (\text{B.7})$$

Further, let $\mathbb{U}_p \subset (0, 1)$ denote a neighborhood of the point $p \in (0, 1)$ in which the quantile function F^{-1} is continuously differentiable (for the proof of Theorem 3.1), or Hölder continuous of order $\gamma \in (0, 1]$ (for the proof of Theorem 3.2). In both cases, the following inequality holds (e.g., Shorack and Wellner, 1986, pp. 453–454):

$$\mathbf{P}(U_{[np]+1:n} \notin \mathbb{U}_p) \leq e^{-cn}, \quad (\text{B.8})$$

for some constant $c > 0$. The inequality above will be used repeatedly throughout the proofs of the two theorems.

Proof of Theorem 3.1. Recall the relationship noted in (B.7). In what follows, we will prove that

$$\mathbf{E}(R_{n,1}) = o(n^{-1}) \quad (\text{B.9})$$

and that

$$\mathbf{E}(R_{n,2}) = \frac{p}{2nf(\xi_p)} + o(n^{-1}), \quad (\text{B.10})$$

which together yield the desired asymptotic expansion in (3.6).

To do so, define the following shorthand notation: $g(u) := \frac{d}{du}(F^{-1}(u)) = 1/f(F^{-1}(u))$ for $u \in \mathbb{U}_p$. We fix $\epsilon_1 > 0$ and $\delta > 0$ such that $I_\delta := (p - \delta, p + \delta) \subset \mathbb{U}_p$ and $|g(u) - g(v)| \leq \epsilon_1$ for all $u, v \in I_\delta$. To prove (B.9), we write

$$\mathbf{E}(R_{n,1}) = \mathbf{E}_1 + \mathbf{E}_2, \quad (\text{B.11})$$

where

$$\mathbf{E}_1 := \mathbf{E}[R_{n,1} \mathbb{1}_{I_\delta}(U_{[np]+1:n})], \quad \mathbf{E}_2 := \mathbf{E}[R_{n,1} \mathbb{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n})].$$

Hereafter, $\mathbb{1}_D(\cdot)$ denotes the indicator function of the set D . Let us first study $\mathbf{E}_1$. The following relationships hold:

$$\begin{aligned} \mathbf{E}_1 &= \frac{np - [np]}{n(1 - p)} \mathbf{E} \left[g(\zeta) (U_{[np]+1:n} - p) \mathbb{1}_{I_\delta}(U_{[np]+1:n}) \right] \\ &= \frac{np - [np]}{n(1 - p)} \mathbf{E} \left[\left(g(p) (U_{[np]+1:n} - p) + (g(\zeta) - g(p)) (U_{[np]+1:n} - p) \right) \mathbb{1}_{I_\delta}(U_{[np]+1:n}) \right], \end{aligned} \quad (\text{B.12})$$

where ζ is a point lying between p and $U_{[np]+1:n}$.

Moreover, consider the following decomposition:

$$\mathbf{E} \left[\left(g(p) (U_{[np]+1:n} - p) + (g(\zeta) - g(p)) (U_{[np]+1:n} - p) \right) \mathbb{1}_{I_\delta}(U_{[np]+1:n}) \right] = e_1 + e_2 + e_3. \quad (\text{B.13})$$

For the first component in the above decomposition, we have

$$e_1 := g(p)\mathbf{E}(U_{[np]+1:n} - p) = g(p) \left(\frac{[np] + 1}{n + 1} - p \right),$$

thus it holds that

$$|e_1| \leq g(p) \frac{\max(p, 1 - p)}{n + 1}. \quad (\text{B.14})$$

For the second component which is given by

$$e_2 := -g(p)\mathbf{E} \left[(U_{[np]+1:n} - p)(1 - \mathbf{1}_{I_\delta}(U_{[np]+1:n})) \right],$$

we have

$$|e_2| \leq g(p)\mathbf{E} \left[1 - \mathbf{1}_{I_\delta}(U_{[np]+1:n}) \right] = g(p)\mathbf{P} \left(U_{[np]+1:n} \notin I_\delta \right) \leq g(p) \exp(-cn), \quad (\text{B.15})$$

where the rightmost inequality holds due to (B.8). Here and throughout, c denotes a positive constant that does not depend on n but may vary from line to line.

Now, consider the third component in the decomposition (B.13), which is given by

$$e_3 := \mathbf{E} \left[(g(\zeta) - g(p))(U_{[np]+1:n} - p) \mathbf{1}_{I_\delta}(U_{[np]+1:n}) \right]. \quad (\text{B.16})$$

By the choice of ϵ_1 and δ noted at the beginning of this proof, and using well-known formulas for the variance of order statistics (see, e.g., [Shorack and Wellner, 1986](#)), we can obtain

$$|e_3| \leq \epsilon_1 \mathbf{E} |U_{[np]+1:n} - p| \leq \epsilon_1 \left(\frac{p(1-p)}{n} (1 + O(n^{-1})) \right)^{1/2}. \quad (\text{B.17})$$

Combining (B.12)–(B.17) then yields

$$\mathbf{E}_1 = o(n^{-1}). \quad (\text{B.18})$$

Next, we turn to $\mathbf{E}_2$ in (B.11), which can be expressed as

$$\frac{np - [np]}{n(1-p)} \mathbf{E} \left[(F^{-1}(U_{[np]+1:n}) - F^{-1}(p)) \mathbf{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n}) \right].$$

Let us consider the following decomposition:

$$\mathbf{E} \left[(F^{-1}(U_{[np]+1:n}) - F^{-1}(p)) \mathbf{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n}) \right] = e_4 + e_5, \quad (\text{B.19})$$

where

$$\begin{aligned} e_4 &:= \mathbf{E}[F^{-1}(U_{[np]+1:n})\mathbf{1}_{[0,1]\setminus I_\delta}(U_{[np]+1:n})], \\ e_5 &:= -F^{-1}(p)\mathbf{E}[\mathbf{1}_{[0,1]\setminus I_\delta}(U_{[np]+1:n})]. \end{aligned}$$

For e_4 , it holds that

$$\begin{aligned} |e_4| &\leq \mathbf{E}[|F^{-1}(U_{[np]+1:n})|\mathbf{1}_{[0,1]\setminus I_\delta}(U_{[np]+1:n})] \\ &\leq [\mathbf{E}|F^{-1}(U_{[np]+1:n})|^{1+\nu}]^{\frac{1}{1+\nu}} [\mathbf{E}(\mathbf{1}_{[0,1]\setminus I_\delta}(U_{[np]+1:n}))]^{1-\frac{1}{1+\nu}} \\ &\leq \underbrace{C(\rho)^{\frac{1}{1+\nu}}}_{C_1(\rho,\nu)} \left[\left(\mathbf{E}|X|^\varepsilon \frac{(n+1)^2}{([np]+1)(n-[np])} \right)^{\frac{1+\nu}{\varepsilon}} \right]^{\frac{1}{1+\nu}} [\mathbf{P}(U_{[np]+1:n} \notin I_\delta)]^{1-\frac{1}{1+\nu}} \\ &\leq C_1(\rho,\nu) \left(\mathbf{E}|X|^\varepsilon \frac{(n+1)^2}{([np]+1)(n-[np])} \right)^{\frac{1}{\varepsilon}} \exp\left(-n \frac{c\nu}{1+\nu}\right) \\ &= C_1(\rho,\nu) \left(\frac{\mathbf{E}|X|^\varepsilon}{p(1-p)} (1+O(n^{-1})) \right)^{\frac{1}{\varepsilon}} \exp\left(-n \frac{c\nu}{1+\nu}\right). \end{aligned} \quad (\text{B.20})$$

In the above calculations, the second inequality follows from the Hölder inequality with an arbitrary $\nu > 0$, while the third inequality applies the inequality in (B.4) with $\rho = (1+\nu)/\varepsilon$.

For e_5 , we get

$$|e_5| = |F^{-1}(p)| \mathbf{P}(U_{[np]+1:n} \notin I_\delta) \leq |F^{-1}(p)| \exp(-cn). \quad (\text{B.21})$$

Combining (B.19)–(B.21) then implies

$$\mathbf{E}_2 = o(n^{-1}). \quad (\text{B.22})$$

Collectively, the relationship in (B.9) holds because of (B.18) and (B.22).

So far, we have established (B.9). Now, we proceed to proving (B.10), which is more demanding. Recall that ϵ_1 and δ are defined and fixed at the beginning of the proof of this theorem. We note that if $U_{[np]+1:n} \in I_\delta$, then $U_{i:n} \in I_\delta$ for all $i = ([np] \wedge M_p) + 1, \dots, [np] \vee M_p$. Indeed, let $U_{[np]+1:n} \in I_\delta$. If $M_p > [np]$, then $U_{[np]+1:n} \leq U_{M_p:n} \leq p$ and $(U_{[np]+1:n}, U_{M_p:n}) \subset I_\delta$. If $M_p < [np]$, then $p < U_{M_p+1:n} \leq U_{[np]+1:n}$, and hence, $(U_{M_p+1:n}, U_{[np]:n}) \subset I_\delta$. If $M_p = [np]$, then $R_{n,2}$ is zero. In passing, we note that, whenever $M_p \neq [np]$, the following inequality holds:

$$\max_{([np] \wedge M_p) + 1 \leq i \leq ([np] \vee M_p)} |F^{-1}(U_{i:n}) - F^{-1}(p)| \leq |F^{-1}(U_{[np]:n}) - F^{-1}(p)|. \quad (\text{B.23})$$

This inequality will be useful later.

We evaluate $\mathbf{E}(R_{n,2})$ in (B.10) by decomposing it as

$$\mathbf{E}(R_{n,2}) = \mathbf{E}_3 + \mathbf{E}_4, \quad (\text{B.24})$$

where

$$\mathbf{E}_3 := \mathbf{E}[R_{n,2} \mathbf{1}_{I_\delta}(U_{[np]+1:n})], \quad \mathbf{E}_4 := \mathbf{E}[R_{n,2} \mathbf{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n})].$$

We first treat $\mathbf{E}_3$. Consider $R_{n,2}$ on the event $\{U_{[np]+1:n} \in I_\delta\}$, on which $\mathbf{1}_{I_\delta}(U_{[np]+1:n}) = 1$. On this event, we have

$$\begin{aligned} R_{n,2} &= -\frac{\text{sign}(M_p - [np])}{n(1-p)} \sum_{i=(\lfloor np \rfloor \wedge M_p)+1}^{\lfloor np \rfloor \vee M_p} g(p)(U_{i:n} - p) \\ &\quad - \frac{\text{sign}(M_p - [np])}{n(1-p)} \sum_{i=(\lfloor np \rfloor \wedge M_p)+1}^{\lfloor np \rfloor \vee M_p} (g(\zeta_i) - g(p))(U_{i:n} - p) =: R'_{n,2} + R''_{n,2}, \end{aligned} \quad (\text{B.25})$$

where each ζ_i lies between p and $U_{i:n}$, $R'_{n,2}$ and $R''_{n,2}$ denote the first and second summation terms in (B.25), respectively. Accordingly, $\mathbf{E}_3$ can be written as

$$\begin{aligned} \mathbf{E}_3 &= \mathbf{E}[(R'_{n,2} + R''_{n,2}) \mathbf{1}_{I_\delta}(U_{[np]+1:n})] \\ &= \mathbf{E}(R'_{n,2}) - \mathbf{E}[R'_{n,2}(1 - \mathbf{1}_{I_\delta}(U_{[np]+1:n}))] + \mathbf{E}[R''_{n,2} \mathbf{1}_{I_\delta}(U_{[np]+1:n})] \\ &=: \mathbf{E}'_3 + e_6 + e_7. \end{aligned} \quad (\text{B.26})$$

We claim that $|e_6 + e_7| = o(n^{-1})$. To see this, for e_6 , we have

$$\begin{aligned} |e_6| &\leq \frac{g(p)}{n(1-p)} \mathbf{E}(|M_p - [np]| (1 - \mathbf{1}_{I_\delta}(U_{[np]+1:n}))) \\ &\leq \frac{g(p)}{n(1-p)} \left(\mathbf{E}(|M_p - [np]|)^2 \mathbf{P}(U_{[np]+1:n} \notin I_\delta) \right)^{1/2} \leq \frac{g(p)p^{1/2}}{(n(1-p))^{1/2}} \exp(-cn). \end{aligned}$$

For e_7 , we get

$$\begin{aligned} |e_7| &\leq \frac{\sup_{u \in I_\delta} |g(u) - g(p)|}{n(1-p)} \mathbf{E}(|M_p - [np]| |U_{[np]:n} - p|) \\ &\leq \frac{\epsilon_1}{n(1-p)} \left(\mathbf{E}(M_p - [np])^2 \mathbf{E}(U_{[np]:n} - p)^2 \right)^{1/2} \\ &= \frac{\epsilon_1 (np(1-p))^{1/2}}{n(1-p)} \left(\frac{p(1-p)}{n} (1 + O(n^{-1})) \right)^{1/2} = \frac{\epsilon_1 p}{n} + O(n^{-2}), \end{aligned}$$

where we used the inequality in (B.23) and the Cauchy–Schwarz inequality to establish the desired

relationships. Since $\epsilon_1 > 0$ is arbitrary, it follows that $e_7 = o(n^{-1})$.

We now evaluate the first term in (B.26). Recall that

$$\mathbf{E}'_3 = \mathbf{E}(R'_{n,2}) = -\frac{g(p)}{1-p} \mathbf{E}\left(\frac{1}{n} \Sigma_n\right), \quad (\text{B.27})$$

where

$$\Sigma_n := \text{sign}(M_p - [np]) \sum_{i=([np] \wedge M_p)+1}^{[np] \vee M_p} (U_{i:n} - p).$$

To analyze the right-hand side of (B.27), we apply a Bahadur–Kiefer-type representation for uniform sample quantiles. Specifically, as $n \rightarrow \infty$, the following representation holds,

$$\frac{1}{n} \Sigma_n = -\frac{(M_p - np)^2}{2n^2} + r_n, \quad (\text{B.28})$$

where for every $c > 0$, there exists a constant A , independent of n , such that

$$\mathbf{P}(|r_n| > A(\log n/n)^{5/4}) = O(n^{-c}). \quad (\text{B.29})$$

Equation (B.28) represents a particular application of Corollary 3.1 of [Gribkova and Helmers \(2006\)](#) for the case $k = [np]$. Note that while Corollary 3.1 of [Gribkova and Helmers \(2006\)](#) deals with arbitrary distributions and thus requires a smoothness condition on the density, the uniform density is identically equal to 1 on $[0, 1]$ and is therefore inherently smooth. We also refer to Theorem 2.2 and Corollary 2.2 of [Gribkova and Helmers \(2012\)](#) for related yet more general results.

Since $\mathbf{E}(M_p - np)^2 = np(1-p)$, taking expectations in (B.28) gives

$$\mathbf{E}\left(\frac{1}{n} \Sigma_n\right) = -\frac{p(1-p)}{2n} + \mathbf{E}(r_n), \quad (\text{B.30})$$

and hence

$$\mathbf{E}'_3 = \frac{pg(p)}{2n} - \frac{g(p)}{1-p} \mathbf{E}(r_n). \quad (\text{B.31})$$

Next, we aim to show that $\mathbf{E}(r_n) = o(n^{-1})$ as $n \rightarrow \infty$. To do so, we first note that Σ_n contains $|M_p - [np]|$ terms, each of absolute value at most 1, so $|\Sigma_n| \leq |M_p - [np]|$. Then, by (B.28),

$$|r_n| \leq \frac{|\Sigma_n|}{n} + \frac{(M_p - np)^2}{2n^2} \leq \frac{|M_p - [np]|}{n} + \frac{(M_p - np)^2}{2n^2} \leq \frac{3}{2}. \quad (\text{B.32})$$

Applying (B.29) with $c = 2$, we find a constant $C > 0$, independent of n , such that $\mathbf{P}(|r_n| > a_n) \leq Cn^{-2}$ for all sufficiently large n , where $a_n := A(\log n/n)^{5/4}$. Since $|r_n| \leq a_n$ on $\{|r_n| \leq a_n\}$ and

$|r_n| \leq 3/2$ everywhere by (B.32), we obtain

$$\begin{aligned} |\mathbf{E}(r_n)| &\leq \mathbf{E}|r_n| = \mathbf{E}(|r_n| \mathbf{1}_{\{|r_n| \leq a_n\}}) + \mathbf{E}(|r_n| \mathbf{1}_{\{|r_n| > a_n\}}) \\ &\leq a_n + \frac{3}{2} \mathbf{P}(|r_n| > a_n) \leq A \frac{\log^{5/4} n}{n^{5/4}} + \frac{3C}{2n^2} \end{aligned} \quad (\text{B.33})$$

for all sufficiently large n . Combining (B.31) and (B.33), we conclude that as $n \rightarrow \infty$,

$$\mathbf{E}'_3 = \frac{p g(p)}{2n} + O\left(\frac{\log^{5/4} n}{n^{5/4}}\right) = \frac{p}{2n f(\xi_p)} + o(n^{-1}). \quad (\text{B.34})$$

To complete the proof, it remains to show that the term $\mathbf{E}_4$ in (B.24) satisfies

$$\mathbf{E}_4 = o(n^{-1}). \quad (\text{B.35})$$

To this end, the following bound is established:

$$\begin{aligned} |\mathbf{E}_4| &\leq \mathbf{E}[|R_{n,2}| \mathbf{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n})] \\ &= \frac{1}{n(1-p)} \mathbf{E} \left[\left| \sum_{i=(\lfloor np \rfloor \wedge M_p)+1}^{\lfloor np \rfloor \vee M_p} (F^{-1}(U_{i:n}) - F^{-1}(p)) \right| \mathbf{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n}) \right] \\ &\leq \frac{1}{1-p} \mathbf{E} \left[\left| F^{-1}(U_{[np]:n}) - F^{-1}(p) \right| \mathbf{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n}) \right] \\ &\leq \frac{1}{1-p} \mathbf{E} \left[\left| F^{-1}(U_{[np]:n}) \right| \mathbf{1}_{[0,1] \setminus I_\delta}(U_{[np]+1:n}) \right] + \frac{|F^{-1}(p)|}{1-p} \mathbf{P}(U_{[np]+1:n} \notin I_\delta), \end{aligned} \quad (\text{B.36})$$

where, in particular, the inequality in (B.23) was used in the third step. The first term on the right-hand side of (B.36) has been bounded analogously in (B.20). Consequently, for an arbitrary $\nu > 0$, the right-hand side of (B.36) is bounded above by

$$\frac{C_1(\rho, \nu)}{1-p} \left(\frac{\mathbf{E}|X|^\varepsilon}{p(1-p)} (1 + O(n^{-1})) \right)^{\frac{1}{\varepsilon}} \exp\left(-n \frac{c\nu}{1+\nu}\right) + \frac{|F^{-1}(p)|}{1-p} \exp(-cn),$$

where $C_1(\rho, \nu) > 0$ is a constant depending only on ν and $\rho = (1+\nu)/\varepsilon$, as defined in (B.20). Since both terms in the above expression decay exponentially fast in n , this bound implies (B.35).

The proof of Theorem 3.1 is finally completed. $\square$

Proof of Theorem 3.2. Recall that, from (B.5)–(B.7), we have

$$-B_n = \mathbf{E}(R_n) = \mathbf{E}(R_{n,1}) + \mathbf{E}(R_{n,2}). \quad (\text{B.37})$$

We shall first show that $\mathbf{E}(R_{n,1})$ is of negligible order relative to the right-hand side of (3.8). To this end, we write

$$\mathbf{E}(R_{n,1}) = \mathbf{E}_1 + \mathbf{E}_2,$$

where

$$\mathbf{E}_1 := \mathbf{E}(R_{n,1} \mathbf{1}_{\mathbb{U}_p}(U_{[np]+1:n})), \quad \mathbf{E}_2 := \mathbf{E}(R_{n,1} \mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{[np]+1:n})).$$

Here, $\mathbb{U}_p$ denotes the neighborhood of p on which F^{-1} is locally Hölder continuous of order $\gamma \in (0, 1]$ with constant c_γ , as specified in (3.7).

For $\mathbf{E}_1$, we get

$$\begin{aligned} |\mathbf{E}_1| &\leq \frac{c_\gamma}{n(1-p)} \mathbf{E}|U_{[np]+1:n} - p|^\gamma \leq \frac{c_\gamma}{n(1-p)} \left(\mathbf{E}(U_{[np]+1:n} - p)^2 \right)^{\gamma/2} \\ &= \frac{c_\gamma}{n(1-p)} \left(\frac{p(1-p)}{n} (1 + o(1)) \right)^{\gamma/2} = \frac{c_\gamma p^{\gamma/2}}{n^{1+\gamma/2} (1-p)^{1-\gamma/2}} (1 + o(1)). \end{aligned}$$

Therefore, $|\mathbf{E}_1| = O(n^{-1-\gamma/2}) = o(n^{-1})$, which is negligible relative to the bound in (3.8).

For $\mathbf{E}_2$, we have

$$\begin{aligned} |\mathbf{E}_2| &= \frac{np - [np]}{n(1-p)} \left| \mathbf{E} \left((F^{-1}(U_{[np]+1:n}) - F^{-1}(p)) \mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{[np]+1:n}) \right) \right| \\ &\leq \frac{1}{n(1-p)} \left(\mathbf{E} \left| F^{-1}(U_{[np]+1:n}) \mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{[np]+1:n}) \right| + |F^{-1}(p)| \mathbf{P}(U_{[np]+1:n} \notin \mathbb{U}_p) \right). \end{aligned}$$

Arguing as in the estimation of e_4 in (B.20), it follows that for any $\nu > 0$,

$$|\mathbf{E}_2| \leq \frac{1}{n(1-p)} \left(C_1(\rho, \nu) \left(\frac{\mathbf{E}|X|^\varepsilon}{p(1-p)} (1 + O(n^{-1})) \right)^{\frac{1}{\varepsilon}} \exp \left(-n \frac{c\nu}{1+\nu} \right) + |F^{-1}(p)| e^{-cn} \right),$$

where $\rho = (1 + \nu)/\varepsilon$. Since both terms in the above expression decay exponentially fast in n , we can conclude that $\mathbf{E}_2 = o(n^{-1})$, and hence $\mathbf{E}_2$ is negligible relative to the bound in (3.8).

It remains to treat $\mathbf{E}(R_{n,2})$ in (B.37). We decompose it as

$$\mathbf{E}(R_{n,2}) = \mathbf{E}_3 + \mathbf{E}_4,$$

where

$$\mathbf{E}_3 := \mathbf{E}(R_{n,2} \mathbf{1}_{\mathbb{U}_p}(U_{[np]+1:n})), \quad \mathbf{E}_4 := \mathbf{E}(R_{n,2} \mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{[np]+1:n})).$$

We first consider the term $\mathbf{E}_3$. Note that all terms in the sum defining $R_{n,2}$ in (B.6) have the same sign, either all positive or all negative, so the absolute value of the sum equals the sum of the absolute values. By the local Hölder continuity of F^{-1} , we can obtain

$$|\mathbf{E}_3| \leq \frac{c_\gamma}{n(1-p)} \mathbf{E} \left(\sum_{i=(\lfloor np \rfloor \wedge M_p)+1}^{\lfloor np \rfloor \vee M_p} |U_{i:n} - p|^\gamma \right). \quad (\text{B.38})$$

Moreover, by (B.23), the largest term in the sum in (B.38) is bounded above by $|U_{\lfloor np \rfloor:n} - p|^\gamma$. Consequently, the right-hand side of (B.38) is bounded by

$$\begin{aligned} & \frac{c_\gamma}{n(1-p)} \mathbf{E} \left(|M_p - \lfloor np \rfloor| |U_{\lfloor np \rfloor:n} - p|^\gamma \right) \\ & \leq \frac{c_\gamma}{n(1-p)} \mathbf{E} \left((|M_p - np| + 1) |U_{\lfloor np \rfloor:n} - p|^\gamma \right) \\ & \leq \frac{c_\gamma}{n(1-p)} \left[\mathbf{E} (|M_p - np| + 1)^2 \mathbf{E} |U_{\lfloor np \rfloor:n} - p|^{2\gamma} \right]^{1/2} \\ & \leq \frac{c_\gamma}{n(1-p)} \left[np(1-p) + 2\mathbf{E} |M_p - np| + 1 \right]^{1/2} \left[(\mathbf{E} (U_{\lfloor np \rfloor:n} - p)^2)^{2\gamma/2} \right]^{1/2} \\ & \leq \frac{c_\gamma}{n(1-p)} \left[np(1-p) + 2(np(1-p))^{1/2} + 1 \right]^{1/2} \left[\frac{p(1-p)}{n} (1 + O(n^{-1})) \right]^{\gamma/2} \\ & = \frac{c_\gamma}{n(1-p)} \left[np(1-p) (1 + O(n^{-1/2})) \right]^{1/2} \left[\frac{p(1-p)}{n} (1 + O(n^{-1})) \right]^{\gamma/2} \\ & = \frac{c_\gamma}{n(1-p)} \left[np(1-p) \right]^{1/2} (1 + O(n^{-1/2})) \left[\frac{p(1-p)}{n} \right]^{\gamma/2} (1 + O(n^{-1})) \\ & = \frac{c_\gamma p^{\frac{1+\gamma}{2}}}{n^{\frac{1+\gamma}{2}} (1-p)^{\frac{1-\gamma}{2}}} (1 + O(n^{-1/2})). \end{aligned} \quad (\text{B.39})$$

The above result leads exactly to the expression of the bound (3.8) in Theorem 3.2.

To complete the proof, it remains to show that the term $\mathbf{E}_4$ is of negligible order relative to the bound on the right-hand side of (B.39). Proceeding as before, but now taking into account the indicator $\mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{\lfloor np \rfloor+1:n})$, we obtain

$$\begin{aligned} |\mathbf{E}_4| & \leq \frac{1}{n(1-p)} \mathbf{E} \left(|M_p - \lfloor np \rfloor| |F^{-1}(U_{\lfloor np \rfloor:n}) - F^{-1}(p)| \mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{\lfloor np \rfloor+1:n}) \right) \\ & \leq \frac{1}{1-p} \mathbf{E} \left(|F^{-1}(U_{\lfloor np \rfloor:n}) - F^{-1}(p)| \mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{\lfloor np \rfloor+1:n}) \right) \\ & \leq \frac{1}{1-p} \mathbf{E} \left(|F^{-1}(U_{\lfloor np \rfloor:n})| \mathbf{1}_{[0,1] \setminus \mathbb{U}_p}(U_{\lfloor np \rfloor+1:n}) \right) + \frac{|F^{-1}(p)|}{1-p} \mathbf{P}(U_{\lfloor np \rfloor+1:n} \notin \mathbb{U}_p) \\ & \leq \frac{C_1(\rho, \nu)}{1-p} \left(\frac{\mathbf{E} |X|^\varepsilon}{p(1-p)} (1 + O(n^{-1})) \right)^{\frac{1}{\varepsilon}} \exp \left(-n \frac{c\nu}{1+\nu} \right) + \frac{|F^{-1}(p)|}{1-p} \exp(-cn), \end{aligned}$$

for each $\nu > 0$ and $\rho = (1 + \nu)/\varepsilon$; see a similar derivation in (B.20). This implies that $\mathbf{E}_4$ is of exponentially small order and therefore negligible relative to the bound in (B.39). This completes the proof of Theorem 3.2. $\square$